ETSI TS 103 650-1 V1.2.1 (2022-08)

EMTEL;
Testing - Conformance test specifications for core elements for network independent access to emergency services (NG112);
Part 1: Protocol Implementation Conformance Statement (PICS), Test Suite Structure and Test Purposes (TSS & TP)

Reference
RTS/EMTEL-00060-1

Keywords
conformance, emergency, emergency services,
interoperability, testing

ETSI

650 Route des Lucioles
F-06921 Sophia Antipolis Cedex - FRANCE

Tel.: +33 4 92 94 42 00 Fax: +33 4 93 65 47 16

Siret N° 348 623 562 00017 - APE 7112B
Association à but non lucratif enregistrée à la
Sous-Préfecture de Grasse (06) N° w061004871

Important notice

The present document can be downloaded from:
http://www.etsi.org/standards-search

The present document may be made available in electronic versions and/or in print. The content of any electronic and/or print versions of the present document shall not be modified without the prior written authorization of ETSI. In case of any existing or perceived difference in contents between such versions and/or in print, the prevailing version of an ETSI deliverable is the one made publicly available in PDF format at www.etsi.org/deliver.

Users of the present document should be aware that the document may be subject to revision or change of status.
Information on the current status of this and other ETSI documents is available at
https://portal.etsi.org/TB/ETSIDeliverableStatus.aspx

If you find errors in the present document, please send your comment to one of the following services:
https://portal.etsi.org/People/CommiteeSupportStaff.aspx

If you find a security vulnerability in the present document, please report it through our
Coordinated Vulnerability Disclosure Program:
https://www.etsi.org/standards/coordinated-vulnerability-disclosure

Notice of disclaimer & limitation of liability

The information provided in the present deliverable is directed solely to professionals who have the appropriate degree of experience to understand and interpret its content in accordance with generally accepted engineering or other professional standard and applicable regulations.
No recommendation as to products and services or vendors is made or should be implied.
No representation or warranty is made that this deliverable is technically accurate or sufficient or conforms to any law and/or governmental rule and/or regulation and further, no representation or warranty is made of merchantability or fitness for any particular purpose or against infringement of intellectual property rights.
In no event shall ETSI be held liable for loss of profits or any other incidental or consequential damages.

Any software contained in this deliverable is provided "AS IS" with no warranties, express or implied, including but not limited to, the warranties of merchantability, fitness for a particular purpose and non-infringement of intellectual property rights and ETSI shall not be held liable in any event for any damages whatsoever (including, without limitation, damages for loss of profits, business interruption, loss of information, or any other pecuniary loss) arising out of or related to the use of or inability to use the software.

Copyright Notification

No part may be reproduced or utilized in any form or by any means, electronic or mechanical, including photocopying and microfilm except as authorized by written permission of ETSI.
The content of the PDF version shall not be modified without the written authorization of ETSI.
The copyright and the foregoing restriction extend to reproduction in all media.

© ETSI 2022.
All rights reserved.

Contents

Intellectual Property Rights 5
Foreword 5
Modal verbs terminology 5
1 Scope 6
2 References 6
2.1 Normative references 6
2.2 Informative references 7
3 Definition of terms, symbols and abbreviations 7
3.1 Terms 7
3.2 Symbols 7
3.3 Abbreviations 7
4 Protocol Implementation Conformance Statement (PICS) 8
4.1 Introduction 8
4.2 Entities 8
4.3 LIS features 9
4.4 ESRP features 9
4.5 ECRF features 10
4.6 PSAP features 10
4.7 BCF features 10
4.8 Mnemonics for PICS reference 11
5 Test Configurations 12
5.1 LIS Test Configurations 12
5.1.1 CFG_LIS_01 12
5.2 ESRP Test Configurations 12
5.2.1 CFG_ESRP_01 12
5.3 ECRF Test Configurations 12
5.3.1 CFG_ECRF_01 12
5.4 PSAP Test Configurations 13
5.4.1 CFG_PSAP_01 13
5.5 BCF Test Configurations 13
5.5.1 CFG_BCF_01 13
6 Test Suite Structure (TSS) 13
6.1 Structure for NG112 tests 13
6.2 Test groups 13
6.2.1 Root 13
6.2.2 Test group 13
6.2.3 Test sub-group 13
6.2.4 Categories 13
7 Test Purposes (TP) 14
7.1 Introduction 14
7.1.1 TP definition conventions 14
7.1.2 TP Identifier naming conventions 15
7.1.3 Rules for the behaviour description 15
7.1.4 Pre-defined initial conditions 16
7.1.4.1 ESRP initial conditions 16
7.1.4.2 SIP initial conditions 18
7.1.5 Sources of TP definitions 19
7.1.6 Mnemonics for PICS reference 19
7.2 Test purposes 19
7.2.1 LIS 19
7.2.1.1 HELD 19
7.2.1.2 Error 23

7.2.1.3 Subscription/Notifications 24
7.2.1.4 Advanced Mobile Location 28
7.2.1.5 Security 30
7.2.2 ESRP 31
7.2.2.1 FindService 31
7.2.2.2 Fixed Target 34
7.2.2.3 SIP REGISTER Target 35
7.2.2.4 'Dequeue Registration Target 35
7.2.2.5 QueueState 36
7.2.2.6 Subscriptions/Notifications 39
7.2.2.7 Security 42
7.2.2.8 Headers 43
7.2.3 ECRF 44
7.2.3.1 LoST 44
7.2.3.2 Errors 53
7.2.3.3 Subscriptions/Notifications 54
7.2.3.4 Security 56
7.2.4 PSAP 57
7.2.4.1 Invite 57
7.2.4.2 Ack 61
7.2.4.3 Bye 62
7.2.4.4 Message 63
7.2.4.5 Options 64
7.2.4.6 Cancel 64
7.2.4.7 Info 65
7.2.4.8 Dequeue registration/deregistration 65
7.2.4.9 HELD 67
7.2.4.10 LoST 68
7.2.4.11 Security 69
7.2.5 BCF 69
7.2.5.1 Invite 69
7.2.5.2 Message 72
7.2.5.3 Security 74
History 75

Intellectual Property Rights

Essential patents

IPRs essential or potentially essential to normative deliverables may have been declared to ETSI. The declarations pertaining to these essential IPRs, if any, are publicly available for **ETSI members and non-members**, and can be found in ETSI SR 000 314: *"Intellectual Property Rights (IPRs); Essential, or potentially Essential, IPRs notified to ETSI in respect of ETSI standards"*, which is available from the ETSI Secretariat. Latest updates are available on the ETSI Web server (https://ipr.etsi.org/).

Pursuant to the ETSI Directives including the ETSI IPR Policy, no investigation regarding the essentiality of IPRs, including IPR searches, has been carried out by ETSI. No guarantee can be given as to the existence of other IPRs not referenced in ETSI SR 000 314 (or the updates on the ETSI Web server) which are, or may be, or may become, essential to the present document.

Trademarks

The present document may include trademarks and/or tradenames which are asserted and/or registered by their owners. ETSI claims no ownership of these except for any which are indicated as being the property of ETSI, and conveys no right to use or reproduce any trademark and/or tradename. Mention of those trademarks in the present document does not constitute an endorsement by ETSI of products, services or organizations associated with those trademarks.

DECT™, **PLUGTESTS™**, **UMTS™** and the ETSI logo are trademarks of ETSI registered for the benefit of its Members. **3GPP™** and **LTE™** are trademarks of ETSI registered for the benefit of its Members and of the 3GPP Organizational Partners. **oneM2M™** logo is a trademark of ETSI registered for the benefit of its Members and of the oneM2M Partners. **GSM**® and the GSM logo are trademarks registered and owned by the GSM Association.

Foreword

This Technical Specification (TS) has been produced by ETSI Special Committee Emergency Communications (EMTEL).

The present document is part 1 of a multi-part deliverable covering Conformance test specifications for core elements for network independent access to emergency services (NG112), as identified below:

Part 1: **"Protocol Implementation Conformance Statement (PICS), Test Suite Structure and Test Purposes (TSS & TP)";**

Part 2: "Abstract Test Suite (ATS) and Protocol Implementation eXtra Information for Testing (PIXIT)".

Modal verbs terminology

In the present document "**shall**", "**shall not**", "**should**", "**should not**", "**may**", "**need not**", "**will**", "**will not**", "**can**" and "**cannot**" are to be interpreted as described in clause 3.2 of the ETSI Drafting Rules (Verbal forms for the expression of provisions).

"**must**" and "**must not**" are **NOT** allowed in ETSI deliverables except when used in direct citation.

1 Scope

The present document provides the Protocol Implementation Conformance Statement (PICS) and Test Suite Structure and Test Purposes (TSS & TP) for core elements for network independent access to emergency services (NG112) as defined in standards listed in clause 2.1 of the present document.

2 References

2.1 Normative references

References are either specific (identified by date of publication and/or edition number or version number) or non-specific. For specific references, only the cited version applies. For non-specific references, the latest version of the referenced document (including any amendments) applies.

Referenced documents which are not found to be publicly available in the expected location might be found at https://docbox.etsi.org/Reference/.

NOTE: While any hyperlinks included in this clause were valid at the time of publication, ETSI cannot guarantee their long term validity.

The following referenced documents are necessary for the application of the present document.

[1] ETSI TS 103 479 (V1.2.1): "Emergency Communications (EMTEL); Core elements for network independent access to emergency services".

[2] Void.

[3] IETF RFC 5985: "HTTP-Enabled Location Delivery (HELD)".

NOTE: Available at https://tools.ietf.org/html/IETF RFC 5985.

[4] IETF RFC 6753: "A Location Dereference Protocol Using HTTP-Enabled Location Delivery (HELD)".

NOTE: Available at https://tools.ietf.org/html/rfc6753.

[5] IETF RFC 5222: "LoST: A Location-to-Service Translation Protocol".

NOTE: Available at https://tools.ietf.org/html/IETF RFC 5222.

[6] IETF RFC 3261: "SIP: Session Initiation Protocol".

NOTE: Available at https://tools.ietf.org/html/IETF RFC 3261.

[7] IETF RFC 5031: "A Uniform Resource Name (URN) for Emergency and Other Well-Known Services".

NOTE: Available at https://tools.ietf.org/html/rfc5031.

[8] IETF RFC 5491: "GEOPRIV Presence Information Data Format Location Object (PIDF-LO) Usage Clarification, Considerations, and Recommendations".

NOTE: Available at https://tools.ietf.org/html/rfc5491.

[9] Void.

[10] IETF RFC 6442: "Location Conveyance for the Session Initiation Protocol".

NOTE: Available at https://datatracker.ietf.org/doc/html/rfc6442.

[11] ETSI TS 103 698 (V1.1.1): "Emergency Communications (EMTEL); Lightweight Messaging Protocol for Emergency Service Accessibility (LMPE)".

2.2 Informative references

References are either specific (identified by date of publication and/or edition number or version number) or non-specific. For specific references, only the cited version applies. For non-specific references, the latest version of the referenced document (including any amendments) applies.

NOTE: While any hyperlinks included in this clause were valid at the time of publication, ETSI cannot guarantee their long term validity.

The following referenced documents are not necessary for the application of the present document but they assist the user with regard to a particular subject area.

[i.1] ISO/IEC 9646-1 (1994): "Information technology -- Open Systems Interconnection -- Conformance testing methodology and framework -- Part 1: General concepts".

[i.2] ISO/IEC 9646-7 (1995): "Information technology -- Open Systems Interconnection -- Conformance testing methodology and framework -- Part 7: Implementation Conformance Statements".

[i.3] EENA: "Next Generation 112 Long Term Definition", Version 1.1, March 2013.

NOTE: Available at https://eena.org/document/ng112-long-term-definition-standard-for-emergency-services/.

3 Definition of terms, symbols and abbreviations

3.1 Terms

For the purposes of the present document, the terms given in ISO/IEC 9646-1 [i.1] and ISO/IEC 9646-7 [i.2] apply.

3.2 Symbols

Void.

3.3 Abbreviations

For the purposes of the present document, the following abbreviations apply:

ACK	Acknowledge
AML	Advanced Mobile Location
ATS	Abstract Test Suite
BCF	Border Control Function
BI	Invalid Behavior
BV	Valid Behaviour
ECRF	Emergency Call. Routing Function
EENA	European Emergency Number Association
EES	European Emergency Services
ESRP	Emergency Service Routing Proxy
HELD	HTTP Enabled Location Discovery
HTTP	Hyper-text Transfer Protocol
ICS	Implementation Conformance Statement
ID	Identity
IM	Instant Messaging
ISO	International Organization for Standardization
IUT	Implementation Under Test
LI	Location Information
LIS	Location Information Server
LO	Location Object
LoST	Location to Service Translation

LTD	Long Term Definition
PICS	Protocol Implementation Conformance Statement
PIDF	Presence Information Data Format
PIXIT	Protocol Implementation eXtra Information for Testing
PSAP	Public Safety Answer Point
SDP	Session Description Protocol
SIP	Session Initiation Protocol
SMS	Short Message Service
TCP	Transmission Control Protocol
TLS	Transport Layer Security
TP	Test Purposes
TS	Test Suite
TSS	Test Suite Structure
UDP	User Datagram Protocol
URI	Uniform Resource Identifier
URN	Universal Resource Name

4 Protocol Implementation Conformance Statement (PICS)

4.1 Introduction

The purpose of a PICS is to identify those standardized functions which an IUT shall support, those which are optional and those which are conditional on the presence of other functions. It helps to identify which functions an IUT will support when performing conformance testing. It is possible that with different choices in an ICS proforma, several different sets of TPs will be necessary.

In the following clauses assessments are made on whether requirements, features, components and other capabilities are required according to a referenced standard and in order to achieve compliance. This assessment provides the following options:

m	mandatory - the capability is required to be supported.
o	optional - the capability may, or may not, be supported.
c.i	conditional - the requirement on the capability ("m", "o", "x" or "n/a") depends on the support of other optional or conditional items. "i" is an integer identifying a unique conditional status expression which is defined immediately following the table.
n/a	not applicable - in the given context, it is not possible to use the capability.
x	prohibited (excluded) - there is a requirement not to use this capability in the given context.
o.i	qualified optional - for mutually exclusive or selectable options from a set: "i" is an integer which identifies a unique group of related optional items and the logic of their selection which is defined immediately following the table.

4.2 Entities

Table 1: Entities

Item	Name of field	Reference	Status	Support
1	LIS	ETSI TS 103 479 [1], clause 5.5	o.1	
2	ESRP	ETSI TS 103 479 [1], clause 5.2	o.1	
3	ECRF	ETSI TS 103 479 [1], clause 5.3	o.1	
4	PSAP	ETSI TS 103 479 [1], clause 5.4	o.1	
5	BCF	ETSI TS 103 479 [1], clause 5.1	o.1	
o.1: At least one of the items shall be supported.				

4.3 LIS features

Table 2: LIS features

Prerequisite: Table 1/1				
Item	**Name of field**	**Reference**	**Status**	**Support**
1	HELD by Value	IETF RFC 5985 [3], clause 4.2	m	
2	HELD by Reference	IETF RFC 5985 [3], clause 4.3	m	
3	HELD by Reference GET	ETSI TS 103 479 [1], clause 8	o	
4	Does the IUT support Point	IETF RFC 5985 [3], clause 6.2 IETF RFC 5491 [8], clause 5.2.1	m	
5	Does the IUT support Circle	IETF RFC 5985 [3], clause 6.2 IETF RFC 5491 [8], clause 5.2.3	m	
6	Does the IUT support Civic Address	IETF RFC 5985 [3], clause 6.2 IETF RFC 5491 [8], clause 3.2	m	
7	Does the IUT support SIP Subscriptions	ETSI TS 103 479 [1], clause 5.5	o	
8	Does the IUT support subscriptions to Element State	ETSI TS 103 479 [1], clause 5.5	o	
9	Does the IUT support subscriptions to Service State	ETSI TS 103 479 [1], clause 5.5	o	
10	Does the IUT support subscriptions to Security Posture	ETSI TS 103 479 [1], clause 5.5	o	
11	Does the IUT support AML via HTTP	ETSI TS 103 479 [1], clause 5.5	o	
12	Does the IUT support AML via Data SMS	ETSI TS 103 479 [1], clause 5.5	o	
13	Does the IUT support AML via Text SMS	ETSI TS 103 479 [1], clause 5.5	o	
14	Does the IUT support mutual TLS authentication	ETSI TS 103 479 [1], clause 6.1.1	m	

4.4 ESRP features

Table 3: ESRP features

Prerequisite: Table 1/2				
Item	**Name of field**	**Reference**	**Status**	**Support**
1	Does the IUT support SIP SUBSCRIPTION		o.1	
2	Does the IUT support dequeue registration		o.1	
3	Does the IUT support fixed target		o.1	
4	Does the IUT support queue state		o	
5	Does the IUT support security posture		o	
6	Does the IUT support element stats		o	
7	Does the IUT support service state		o	
8	Does the IUT support LoST protocol	ETSI TS 103 479 [1], clause 5.2.2	m	
9	Does the IUT support LoST protocol with redirection	ETSI TS 103 479 [1], clause 5.2.2	o	
10	Does the IUT support HELD protocol	ETSI TS 103 479 [1], clause 5.2.2	o	
11	Does the IUT support HELP protocol with location reference	ETSI TS 103 479 [1], clause 5.2.2	o	
12	Does the IUT support TLS mutual authentication	ETSI TS 103 479 [1], clause 5.5	o	
13	Does the IUT support adding Call Identifier	ETSI TS 103 479 [1], clause 6.1.2.7	m	
14	Does the IUT support adding Incident Identifier	ETSI TS 103 479 [1], clause 6.1.2.7	m	
o.1: At least one of the items shall be supported.				

4.5 ECRF features

Table 4: ECRF features

Prerequisite: Table 1/3				
Item	**Name of field**	**Reference**	**Status**	**Support**
1	Does the IUT support Find Service	IETF RFC 5222 [5], clause 14	m	
2	Does the IUT support List Services	ETSI TS 103 479 [1], clause 6.4	m	
3	Does the IUT support List Services by Location			
4	Does the IUT support Point	IETF RFC 5222 [5], clause 12.2 IETF RFC 5491 [8], clause 5.2.1	m	
5	Does the IUT support Circle	IETF RFC 5222 [5], clause 12.2 IETF RFC 5491 [8], clause 5.2.3	m	
6	Does the IUT support Ellipse	IETF RFC 5222 [5], clause 12.2 IETF RFC 5491 [8], clause 5.2.3	o	
7	Does the IUT support Arcband	IETF RFC 5222 [5], clause 8.2 IETF RFC 5491 [8], clause 3.2	o	
8	Does the IUT support Polygon	IETF RFC 5222 [5], clause 12.2 IETF RFC 5491 [8], clause 5.2.3	o	
9	Does the IUT support Loop detection	IETF RFC 5222 [5], clause 6	m	
10	Does the IUT support Redirection	IETF RFC 5222 [5], clause 8.3.3	m	
11	Does the IUT support subscriptions to Element State	ETSI TS 103 479 [1], clause 5.3	o	
12	Does the IUT support subscriptions to Security Posture	ETSI TS 103 479 [1], clause 5.3	o	
13	Does the IUT support subscriptions to Service State	ETSI TS 103 479 [1], clause 5.3	o	
14	Does the IUT support TLS mutual authentication	ETSI TS 103 479 [1], clause 5.5	o	

4.6 PSAP features

Table 5: PSAP features

Prerequisite: Table 1/4				
Item	**Name of field**	**Reference**	**Status**	**Support**
1	PSAP service	ETSI TS 103 479 [1], clause 5.4	m	
2	UDP handling	ETSI TS 103 479 [1], clause 6.1.1	o.1	
3	TCP handling	ETSI TS 103 479 [1], clause 6.1.1	o.1	
4	Does the IUT support Dequeue registration/deregistration	ETSI TS 103 479 [1], clause 6.2.1	o	
5	Does the IUT support HELD protocol	ETSI TS 103 479 [1], clause 6.5.1	o	
6	Does IUT support LoST protocol	ETSI TS 103 479 [1], clause 6.4.1	o	
7	Does the IUT support TLS mutual authentication	ETSI TS 103 479 [1], clause 5.5	o	
o.1: At least one of the items shall be supported.				

4.7 BCF features

Table 6: BCF features

Prerequisite: Table 1/5				
Item	**Name of field**	**Reference**	**Status**	**Support**
1	BCF service	ETSI TS 103 479 [1], clause 5.1	m	
2	UDP handling	ETSI TS 103 479 [1], clause 6.1.1	o.1	
3	TCP handling	ETSI TS 103 479 [1], clause 6.1.1	o.1	
4	Does the IUT support TLS mutual authentication	ETSI TS 103 479 [1], clause 5.5		
o.1: At least one of the items shall be supported.				

4.8 Mnemonics for PICS reference

To avoid an update of all related documents when the PICS document is changed, the table below introduces mnemonic names and the correspondence with the PICS item number.

Table 7: Mnemonics for PICS reference

Mnemonic	PICS item
PICS_LIS_HELD_BY_VALUE	Table 2/1
PICS_LIS_HELD_BY_REFERENCE	Table 2/2
PICS_LIS_HELD_BY_REFERENCE_GET	Table 2/3
PICS_LIS_GEOMETRY_POINT	Table 2/4
PICS_LIS_GEOMETRY_CIRCLE	Table 2/5
PICS_LIS_CIVIC	Table 2/6
PICS_LIS_SIP_SUBSCRIPTION	Table 2/7
PICS_LIS_SIP_ELEMENT_STATE	Table 2/8
PICS_LIS_SIP_SERVICE_STATE	Table 2/9
PICS_LIS_SIP_SECURITY_POSTURE	Table 2/10
PICS_LIS_AML_HTTP	Table 2/11
PICS_LIS_AML_DATA_SMS	Table 2/12
PICS_LIS_AML_TEXT_SMS	Table 2/113
PICS_LIS_TLS_AUTHENTICATION	Table 2/14
PICS_ESRP_TARGET_SIP_REGISTER_SUPPORT	Table 3/1
PICS_ESRP_TARGET_DEQUEUE_REGISTRATION_SUPPORT	Table 3/2
PICS_ESRP_FIXED_TARGET_SUPPORT	Table 3/3
PICS_ESRP_QUEUE_STATE_SUPPORT	Table 3/4
PICS_ESRP_SIP_SECURITY_POSTURE	Table 3/5
PICS_ESRP_SIP_ELEMENT_STATE	Table 3/6
PICS_ESRP_SIP_SERVICE_STATE	Table 3/7
PICS_ESRP_LOST	Table 3/8
PICS_ESRP_LOST_REDIRECT	Table 3/9
PICS_ESRP_HELD	Table 3/10
PICS_ESRP_HELD_LOCATION_REFERENCE	Table 3/11
PICS_ESRP_TLS_AUTHENTICATION	Table 3/12
PICS_ESRP_CALL_IDENTIFIER_SUPPORT	Table 3/13
PICS_ESRP_INCIDENT_IDENTIFIER_SUPPORT	Table 3/14
PICS_ECRF_FIND_SERVICE	Table 4/1
PICS_ECRF_LIST_SERVICES	Table 4/2
PICS_ECRF_LIST_SERVICES_BY_LOCATION	Table 4/3
PICS_ECRF_GEOMETRY_POINT	Table 4/4
PICS_ECRF_GEOMETRY_CIRCLE	Table 4/5
PICS_ECRF_GEOMETRY_ELLIPSE	Table 4/6
PICS_ECRF_GEOMETRY_ARCBAND	Table 4/7
PICS_ECRF_GEOMETRY_POLYGON	Table 4/8
PICS_ECRF_LOOP_DETECTION	Table 4/9
PICS_ECRF_REDIRECT	Table 4/10
PICS_ECRF_SIP_ELEMENT_STATE	Table 4/11
PICS_ECRF_SIP_SECURITY_POSTURE	Table 4/12
PICS_ECRF_SIP_SERVICE_STATE	Table 4/13
PICS_ECRF_TLS_AUTHENTICATION	Table 4/14
PICS_A_SIP_BSC1	Table 5/1
PICS_E_SIP_URN1	Table 5/1
PICS_E_SIP_URN2	Table 5/1
PICS_E_SIP_URN3	Table 5/1
PICS_PSAP_B_SDP_ULA1	Table 5/1
PICS_PSAP_B_SDP_ALA1	Table 5/1
PICS_PSAP_S_SIP_BYE1	Table 5/1
PICS_M_SIP_URN1	Table 5/1
PICS_E_SIP_OPT1	Table 5/1
PICS_E_SIP_ NO_REGISTRATION	Table 5/1
PICS_PSAP_S_SIP_REGISTRATION	Table 5/1
PICS_PSAP_S_SIP_UDP1	Table 5/2
PICS_PSAP_S_SIP_TCP1	Table 5/3
PICS_PSAP_S_HTTP_DEQUEUE_REGISTRATION	Table 5/4
PICS_PSAP_S_HTTP_HELD	Table 5/5

Mnemonic	PICS item
PICS_PSAP_S_HTTP_LOST	Table 5/6
PICS_PSAP_TLS_AUTHENTICATION	Table 5/7
PICS_BCF_S_SIP_TCP1	Table 6/3
PICS_BCF_M_SIP_CALL_INFO	Table 6/1
PICS_BCF_M_SIP_CALL_INFO_INCIDENT_TRACKING_ID	Table 6/1
PICS_BCF_M_SIP_CALL_INFO_CALL_ID	Table 6/1
PICS_BCF_M_SIP_CALL_INFO_SOURCE_ID	Table 6/1
PICS_BCF_S_SIP_TCP1	Table 6/3
PICS_BCF_TLS_AUTHENTICATION	Table 6/4

5 Test Configurations

5.1 LIS Test Configurations

5.1.1 CFG_LIS_01

Figure 1: CFG_LIS_01

5.2 ESRP Test Configurations

5.2.1 CFG_ESRP_01

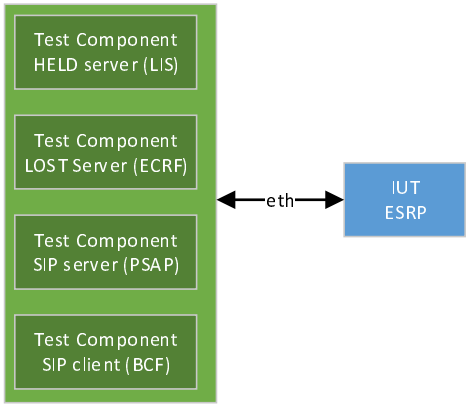

Figure 2: CFG_ESRP_01

5.3 ECRF Test Configurations

5.3.1 CFG_ECRF_01

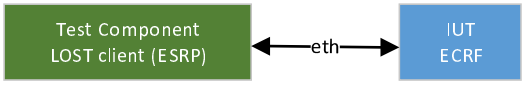

Figure 3: CFG_ECRF_01

5.4 PSAP Test Configurations

5.4.1 CFG_PSAP_01

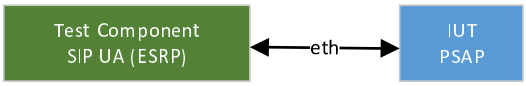

Figure 4: CFG_PSAP_01

5.5 BCF Test Configurations

5.5.1 CFG_BCF_01

Figure 5: CFG_BCF_01

6 Test Suite Structure (TSS)

6.1 Structure for NG112 tests

The test grouping is organized on three (3) levels. The first level is the the reference to the base Document that contains the requirements for the tests. The second level is the Group and identifies the protocol (LIS, ESRP, ECRF, PSAP and BCF) component that is providing the Service or Interface to be tested. The third level is called Subgroup and identifies a set of functionalities.

Moreover, test purposes are identified and categorized by a sequential two-digits number (uniquely assigned upon definition of the each test purpose) and by the type of test performed. The type of test helps quickly identify the type of behaviour that is expected by the IUT in the test purpose.

6.2 Test groups

6.2.1 Root

The root identifies the entities to be tested.

6.2.2 Test group

This level contains the protocols and protocol operations.

6.2.3 Test sub-group

This level identifies the sub categories of each Group.

6.2.4 Categories

This level contains the standard conformance test categories: behaviour for valid, invalid, inopportune events and timers.

7 Test Purposes (TP)

7.1 Introduction

7.1.1 TP definition conventions

The TPs are defined by the rules shown in table 8.

Table 8: TP definition rules

TP Header	
TP ID	The TP ID is a unique identifier. It shall be specified according to the TP naming conventions defined in the above clause.
Test objective	Short description of test purpose objective according to the requirements from the base standard.
Reference	The reference indicates the clauses of the reference standard specifications in which the conformance requirement is expressed.
Config Id	The Config Id references the GeoNetworking configuration selected for this TP.
PICS Selection	Reference to the PICS statement involved for selection of the TP. Contains a Boolean expression.
TP Behaviour	
Initial conditions	The initial conditions define in which initial state the IUT has to be to apply the actual TP. In the corresponding Test Case, when the execution of the initial condition does not succeed, it leads to the assignment of an Inconclusive verdict.
Expected behaviour (TP body)	Definition of the events, which are parts of the TP objective, and the IUT are expected to perform in order to conform to the base specification. In the corresponding Test Case, Pass or Fail verdicts can be assigned there.

7.1.2 TP Identifier naming conventions

The identifier of the TP is built according to table 9.

Table 9: TP naming convention

Identifier:	TP_<root>_<gr>_<sgr>_<x>_<nn>	
	<root> = root	LIS
		ESRP
		ECRF
		PSAP
		BCF
	<gr> = group	HELD
		HELD_BY_REFERENCE
		Error
		LOST
		AML
		FindService
		Fixed_TARGET
		SIP_REGISTER_TARGET
		HTTP_DEQUEUE_REGISTRATION
		QUEUE_STATE
		SIP
		TLS
	<sgr> =sub-group	GET
		PUT
		POST
		INVITE
		REGISTER
		ACK
		BYE
		MESSAGE
		OPTIONS
		CANCEL
		INFO
		SECURITY
	<x> = type of testing	BV
		BI
	<nn> = sequential number	

7.1.3 Rules for the behaviour description

In the TP the following wordings are used:

- "receives": for packets coming from the network to the IUT
- "sends": for packets sent by the IUT to the network
- "forwards": forwards the previously received message to the next hop
- "generates": for internal event generation
- "rejects": connection rejected due to security issue
- "isRequestedToSend": an upper layer requests the IUT to send a packet
- "havingLocationMappingFor": IUT is provisioned with the relevant location data
- "havingReturnedLocationUriFor": IUT returned a locationURI for the relevant location data after a HELD request
- "isConfiguredWith": IUT is configured to use a specific service/parameter set
- "isReachableWith": the IUT is reachable via the specified URI

- "isNotReachable": the PSAP is not reachable
- "havingServiceBoundaryFor": IUT is provisioned with the relevant service boundary
- "serviceMappingFor": IUT is provisioned with the relevant service mapping
- "receivedInitialInviteRequestAndSentLostQueryToEcrf" (for more detail see INIT_CON_1)
- "receivedInitialInviteRequestWithoutLocationAndSentHeldRequestToLisFor" (for more detail see INIT_CON_2)
- "receivedInitialInviteRequestWithLocationReferenceAndSentGetRequestToLisFor" (for more detail see INIT_CON_3)
- "sendsLostQueryToEcrfFor": IUT sends a LoST request to the LIs with the given LOCATION (for more detail see INIT_CON_4)
- "receivesHeldResponseWith": IUT receives a HELD response with the give LOCATION (for more detail see INIT_CON_5)
- "receivesLostResponseWith": IUT receives a LoST response with the give URN (for more detail see INIT_CON_6)
- "receivesLocationResponseWith": IUT receives a Location response with the give LOCATION (for more detail see INIT_CON_7)
- "acceptingIncomingCalls": IUT ready to receive incoming calls
- "establishesIncomingCall": The IUT establishes the incoming call (for more detail see INIT_CON_8)
- "inAnActiveIncomingCall": An incoming call is established (for more detail see INIT_CON_9)
- "isConnected": connection between the IUT and peer is established
- "isDisconnected": connection between the IUT and peer is NOT established
- acceptingIncomingCalls: The IUT is ready to receive incoming calls
- establishesIncomingCall: The IUT establishes the incoming call
- acceptingIM: The IUT is ready to receive IM (Instant messaging)
- isRequestedToGetLocationInfo: The IUT is requested to send a request for location information
- isRequestedToGetRouteEmergencyCalls: The IUT is requested to send a request for route emergency call

7.1.4 Pre-defined initial conditions

7.1.4.1 ESRP initial conditions

INIT_CON_1

```
the IUT entity receives a TCP SIP_INVITE containing
    Request_URI indicating value SERVICE_URN_1,
    Content_Type indicating value "multipart/mixed",
    body containing
        SDP_AND_PIDF_MULTIPART
and the IUT entity sends a POST containing
 Content_type indicating value "application/lost+xml;charset=utf-8",
    body containing
        xmlMessage containing
            version indicating value "1.0",
            element "findService" inNamespace "urn:ietf:params:xml:ns:lost1" containing
                element "location" containing
                    element "Point" inNamespace "http://www.opengis.net/gml" containing
                        attribute "srsName" indicating value "urn:ogc:def:crs:EPSG::4326",
                        element "pos" indicating value LOCATION_1
```

```
                element "service" indicating value SERVICE_URN_1
    to the ECRF entity
```

INIT_CON_2

```
the IUT entity receives a UDP SIP_INVITE containing
    Request_URI indicating value SERVICE_URN_1,
    Content_Type indicating value "application/sdp",
    P-Asserted-Identity indicating value tel:DEVICE_NUMBER,
    body containing
        SDP
and the IUT entity sends a POST containing
 Content_type indicating value "application/lost+xml;charset=utf-8",
    body containing
        xmlMessage containing
          version indicating value "1.0",
          element "locationRequest" inNamespace "urn:ietf:params:xml:ns:geopriv:held" containing
          element "locationType" indicating value "geodetic" containing
            attribute "exact" indicating value "true"
          element "device" inNamespace "urn:ietf:params:xml:ns:geopriv:held:id" containing
            element "uri" inNamespace "urn:ietf:params:xml:ns:geopriv:held:id" indicating value
               DEVICE_NUMBER
    to the LIS entity
```

INIT_CON_3

```
the IUT entity receives a UDP SIP_INVITE containing
    Request_URI indicating value SERVICE_URN_1,
    Content_Type indicating value "application/sdp",
    Geolocation indicating value LOCATION_URI
    body containing
        SDP
and the IUT entity sends a GET to the LOCATION_URI
```

INIT_CON_4

```
the IUT entity sends a POST containing
    Content_type indicating value "application/lost+xml;charset=utf-8",
    body containing
        xmlMessage containing
            version indicating value "1.0",
            element "findService" inNamespace "urn:ietf:params:xml:ns:lost1" containing
                element "location" containing
                    element "Point" inNamespace "http://www.opengis.net/gml" containing
                        attribute "srsName" indicating value "urn:ogc:def:crs:EPSG::4326",
                        element "pos" indicating value LOCATION
                element "service" indicating value SERVICE_URN_1
to the ECRF entity
```

INIT_CON_5

```
the IUT entity receives a httpResponse containing
    Status_Code indicating value "200 OK",
      version indicating value "1.0",
      Content_type indicating value "application/held+xml;charset=utf-8",
      body containing
        xmlMessage containing
          version indicating value "1.0",
          element "locationResponse" inNamespace "urn:ietf:params:xml:ns:geopriv:held" containing
          element "presence" inNamespace "urn:ietf:params:xml:ns:pidf" containing
            attribute "entity" indicating value valid "pres:" uri,
            element "tuple" containing
              attribute "id",
              element "status" containing
                element "geopriv" inNamespace "urn:ietf:params:xml:ns:pidf:geopriv10" containing
                  element "location-info" containing
                  element "Point" inNamespace "http://www.opengis.net/gml" containing
                    attribute "srsName" indicating value "urn:ogc:def:crs:EPSG::4326",
                    element "pos" indicating value LOCATION
   from the LIS entity
```

INIT_CON_6

```
the IUT entity receives a httpResponse containing
    Status_Code indicating value "200 OK",
    version indicating value "1.0",
    Content_type indicating value "application/lost+xml;charset=utf-8",
    body containing
        xmlMessage containing
            version indicating value "1.0",
            element "findServiceResponse" inNamespace "urn:ietf:params:xml:ns:lost1" containing
                element "mapping" containing
                    attribute "source",
                    attribute "sourceId",
                    attribute "lastUpdated",
                    attribute "expires",
                    element "service" indicating value SERVICE_URN_1,
                    element "uri" indicating value [TARGET_URI]
                element "locationUsed"
    from the ECRF entity
```

INIT_CON_7

```
the IUT entity receives a httpResponse containing
  Status_Code indicating value "200 OK",
  version indicating value "1.0",
  Content_type indicating value "application/pidf+xml;charset=utf-8",
  body containing
    xmlMessage containing
    version indicating value "1.0",
    element "presence" inNamespace "urn:ietf:params:xml:ns:pidf" containing
      attribute "entity" indicating value valid "pres:" uri,
      element "tuple" containing
        attribute "id",
        element "status" containing
          element "geopriv" inNamespace "urn:ietf:params:xml:ns:pidf:geopriv10" containing
            element "location-info" containing
              element "Point" inNamespace "http://www.opengis.net/gml" containing
                attribute "srsName" indicating value "urn:ogc:def:crs:EPSG::4326",
                element "pos" indicating value LOCATION
    from the LIS entity
```

INIT_CON_8

```
THEN IUT sends TRYING (optional)
THEN IUT sends RINGING (optional)
THEN IUT sends OK
THEN IUT receives ACK
```

INIT_CON_9

```
WHEN the IUT entity receives a TCP SIP_INVITE containing
  Request_URI indicating value "urn:service:sos.police",
  Content_Type indicating value "multipart/mixed",
  body containing
    SDP_AND_PIDF_MULTIPART
THEN the IUT sends TRYING (optional)
THEN the IUT sends RINGING (optional)
THEN the IUT sends OK
THEN the IUT receives ACK
```

7.1.4.2 SIP initial conditions

SIP_INIT_CON_1

```
    the IUT entity acceptingIncomingCalls
```

SIP_INIT_CON_2

```
    the IUT entity acceptingIM
```

SIP_INIT_CON_3

```
    the IUT entity isRegistered to the ESRP and,
    the IUT entity acceptingIncomingCalls
```

SIP_INIT_CON_4

```
    the IUT entity inAnActiveIncomingCall
```

SIP_INIT_CON_5

```
    the IUT entity isDisconnected
```

7.1.5 Sources of TP definitions

All TPs have been specified according to the referenced standards in clause 2.1.

7.1.6 Mnemonics for PICS reference

The present document makes use of PICS mnemonics defined in table 7.

7.2 Test purposes

7.2.1 LIS

7.2.1.1 HELD

TP Id	TP_LIS_HELD_BV_01
Test Objective	IUT successfully responds with a Point when it receives a location request without location type
Reference	ETSI TS 103 479 [1], clause 6.5.1 EENA Next Generation 112 LTD [i.3], clause 4.10 IETF RFC 5985 [3]
PICS Selection	PICS_LIS_HELD_BY_VALUE and PICS_LIS_GEOMETRY_POINT
Initial Conditions	

with {
the IUT has a Mapping containing
uri indicating value PX_URI,
location indicating value PX_POINT
 	
}	

Expected Behaviour	

ensure that {
when {
the IUT receives an LocationRequest containing
uri corresponding to PX_URI
 	
}
then {
the IUT sends a LocationResponse containing
presence containing
"location-info" carrying PX_POINT
 	
}
}	

TP Id	TP_LIS_HELD_BV_02
Test Objective	IUT successfully responds with a Circle when it receives a location request without location type
Reference	ETSI TS 103 479 [1], clause 6.5.1 EENA Next Generation 112 LTD [i.3], clause 4.10 IETF RFC 5985 [3]
PICS Selection	PICS_LIS_HELD_BY_VALUE and PICS_LIS_GEOMETRY_POINT
Initial Conditions	

```
with {
                the IUT has a Mapping containing
                        uri          indicating value PX_URI,
                           location     indicating value PX_CIRCLE

            }
```

Expected Behaviour

```
ensure that {
                when {
                    the IUT receives an LocationRequest containing
                        uri        corresponding to PX_URI

                  }
                  then {
                    the IUT sends a LocationResponse containing
                        presence containing
                            "location-info"    carrying PX_CIRCLE


                  }
            }
```

TP Id	TP_LIS_HELD_BV_03
Test Objective	IUT successfully responds with a Reference when it receives a location request with location type locationURI and exact attribute
Reference	ETSI TS 103 479 [1], clause 6.5.1 EENA Next Generation 112 LTD [i.3], clause 4.10 IETF RFC 5985 [3]
PICS Selection	PICS_LIS_HELD_BY_REFERENCE_GET and PICS_LIS_GEOMETRY_CIRCLE
Initial Conditions	

```
with {
                the IUT has a Mapping containing
                        uri          indicating value PX_URI,
                           location     indicating value PX_CIRCLE

            }
```

Expected Behaviour

```
ensure that {
                when {
                    the IUT receives an LocationRequest containing
                        uri        corresponding to PX_URI,
                        locationType containing
                            exact    indicating value True,
                            values    indicating value "locationURI"


                  }
                  then {
                    the IUT sends a LocationResponse containing
                        "locationUri"    carrying any URI

                  }
            }
```

TP Id	TP_LIS_HELD_BV_04
Test Objective	IUT successfully responds with a Reference and Circle when it receives a location request with location types locationURI and geodetic and exact attribute
Reference	ETSI TS 103 479 [1], clause 6.5.1 EENA Next Generation 112 LTD [i.3], clause 4.10 IETF RFC 5985 [3]
PICS Selection	PICS_LIS_HELD_BY_REFERENCE_GET and PICS_LIS_GEOMETRY_CIRCLE
Initial Conditions	

```
with {
                the IUT has a Mapping containing
                    uri        set to PX_URI,
                    location set to PX_CIRCLE

            }
```

Expected Behaviour

```
ensure that {
                when {
                    the IUT receives an LocationRequest containing
                        uri        corresponding to PX_URI,
                        locationType containing
                            exact    set to True,
                            values    set to "locationURI,geodetic"


                  }
                  then {
                    the IUT sends a LocationResponse containing
                        locationUri    carrying any URI,
                        presence containing
                            "location-info"    carrying PX_CIRCLE


                  }
            }
```

TP Id	TP_LIS_HELD_BV_05
Test Objective	IUT successfully responds with a Civic when it receives a location request with location type civic
Reference	ETSI TS 103 479 [1], clause 6.5.1 EENA Next Generation 112 LTD [i.3], clause 4.10 IETF RFC 5985 [3]
PICS Selection	PICS_LIS_HELD_BY_VALUE and PICS_LIS_CIVIC
Initial Conditions	

```
with {
                the IUT has a Mapping containing
                    uri        set to PX_URI,
                    address set to PX_ADDRESS

            }
```

Expected Behaviour

```
ensure that {
                when {
                    the IUT receives an LocationRequest containing
                        uri        corresponding to PX_URI,
                        locationType containing
                            exact    set to True,
                            values    set to "civic"


                  }
                  then {
                    the IUT sends a LocationResponse containing
                        presence containing
                            "location-info"    carrying PX_ADDRESS


                  }
            }
```

TP Id	TP_LIS_HELD_BV_06
Test Objective	IUT successfully responds with a Circle when the locationURI is dereferenced
Reference	ETSI TS 103 479 [1], clause 6.5.2 EENA Next Generation 112 LTD [i.3], clause 4.10 IETF RFC 5985 [3]
PICS Selection	PICS_LIS_HELD_BY_REFERENCE_GET and PICS_LIS_GEOMETRY_CIRCLE
Initial Conditions	

```
with {
                the IUT has a Mapping containing
                    uri        set to PX_URI,
                    address set to PX_CIRCLE

                and the IUT receives an LocationRequest containing
                    uri        corresponding to PX_URI,
                    locationType containing
                        exact    indicating value True,
                        values    indicating value "locationURI"


                and the IUT sends a LocationResponse containing
                        "locationUri"    carrying any URI

            }
```

Expected Behaviour

```
ensure that {
                when {
                    the IUT receives an DereferenceRequest containing
                        uri indicating value "from LocationResponse"

                  }
                  then {
                    the IUT sends a PresenceDocument containing
                        "location-info"    carrying PX_CIRCLE

                  }
            }
```

TP Id	TP_LIS_HELD_BV_07
Test Objective	IUT successfully responds with a Circle when the locationURI is dereferenced by HeldRequest
Reference	ETSI TS 103 479 [1], clause 6.5.2 EENA Next Generation 112 LTD [i.3], clause 4.10 IETF RFC 5985 [3]
PICS Selection	PICS_LIS_HELD_BY_REFERENCE_POST and PICS_LIS_GEOMETRY_CIRCLE
Initial Conditions	

with {
the IUT has a Mapping containing
uri set to PX_URI,
address set to PX_CIRCLE
 	
and the IUT receives an LocationRequest containing
uri corresponding to PX_URI,
locationType containing
exact indicating value True,
values indicating value "locationURI"
 	
and the IUT sends a LocationResponse containing
"locationUri" carrying any URI
 	
}	

Expected Behaviour	

ensure that {
when {
the IUT receives an DereferenceHeldRequest containing
uri indicating value "from LocationResponse"
 	
}
then {
the IUT sends a LocationResponse containing
presence containing
"location-info" carrying PX_CIRCLE
 	
}
}	

7.2.1.2 Error

TP Id	TP_LIS_HELD_BI_01
Test Objective	IUT responds with an error response when it receives a location request for an unknown device
Reference	ETSI TS 103 479 [1], clause 6.5.1 EENA Next Generation 112 LTD [i.3], clause 4.10 IETF RFC 5985 [3]
Initial Conditions	

with {
the IUT has no Mapping containing
uri set to PX_URI
 	
}	

Expected Behaviour	

ensure that {
when {
the IUT receives an LocationRequest containing
uri corresponding to PX_URI
 	
}
then {
the IUT sends an ErrorResponse containing
code set to "locationUnknown"
 	
}
}	

TP Id	TP_LIS_HELD_BI_02
Test Objective	IUT responds with an error response when it receives a location request with an unmatched location type
Reference	ETSI TS 103 479 [1], clause 6.5.1 EENA Next Generation 112 LTD [i.3], clause 4.10 IETF RFC 5985 [3]
Initial Conditions	

with {
the IUT has a Mapping containing
uri set to PX_URI,
address set to PX_CIRCLE
 	
}	

Expected Behaviour	

ensure that {
when {
the IUT receives an LocationRequest containing
uri corresponding to PX_URI,
locationType containing
exact indicating value True,
values indicating value "civic"
 	
}
then {
the IUT sends an ErrorResponse containing
code set to "cannotProvideLiType"
 	
}
}	

7.2.1.3 Subscription/Notifications

TP Id	TP_LIS_SIP_BV_01
Test Objective	IUT responds to SIP Subscribe with an 200 OK
Reference	ETSI TS 103 479 [1], clause 6.5.3.2 EENA Next Generation 112 LTD [i.3], clause 4.10 IETF RFC 5985 [3]
PICS Selection	PICS_LIS_SIP_SUBSCRIPTION and PICS_LIS_HELD_BY_REFERENCE_GET and PICS_LIS_GEOMETRY_POINT
Initial Conditions	

with {
the IUT has a Mapping containing
uri indicating value PX_URI,
location indicating value PX_POINT
 	
and the IUT receives an LocationRequest containing
uri corresponding to PX_URI,
locationType containing
exact indicating value True,
values indicating value "locationURI"
 	
and the IUT sends a LocationResponse containing
"locationUri" carrying any URI
 	
}	

Expected Behaviour

ensure that {
when {
the IUT receives a SubscribeRequest containing
"event" set to "presence"
 	
}
then {
the IUT sends an OkResponse
and the IUT sends a Notify containing
presence containing
"location-info" carrying PX_POINT
 	
}
}

TP Id	TP_LIS_SIP_BV_02
Test Objective	IUT responds to SIP Unsubscribe with an 200 OK
Reference	ETSI TS 103 479 [1], clause 6.5.3.2 EENA Next Generation 112 LTD [i.3], clause 4.10 IETF RFC 5985 [3]
PICS Selection	PICS_LIS_SIP_SUBSCRIPTION
Initial Conditions	

with {
the IUT receives a SubscribeRequest containing
"event" set to "presence"
 	
and the IUT sends an OkResponse
and the IUT sends a Notify
}	

Expected Behaviour	

ensure that {
when {
the IUT receives an UnsubscribeRequest
}
then {
the IUT sends an OkResponse
}
}	

TP Id	TP_LIS_SIP_BV_03
Test Objective	IUT sends SIP Notify when location changes
Reference	ETSI TS 103 479 [1], clause 6.5.3.2 EENA Next Generation 112 LTD [i.3], clause 4.10 IETF RFC 5985 [3]
PICS Selection	PICS_LIS_SIP_SUBSCRIPTION
Initial Conditions	

with {
the IUT receives a SubscribeRequest containing
"event" set to "presence"
 	
and the IUT sends an OkResponse
and the IUT sends a Notify
}	

Expected Behaviour	

ensure that {
when {
the IUT detects a LocationChange
}
then {
the IUT sends an Notify containing
Presence
 	
}
}	

TP Id	TP_LIS_SIP_BV_04
Test Objective	IUT responds to SIP Subscribe for Element State with an 200 OK
Reference	ETSI TS 103 479 [1], clause 6.3.4 EENA Next Generation 112 LTD [i.3], clause 4.10 IETF RFC 5985 [3]
PICS Selection	PICS_SIP_SUBSCRIPTION_ELEMENT_STATE
Expected Behaviour	

```
ensure that {
                when {
                    the IUT receives a SubscribeRequest containing
                        "event" set to "emergency-ElementState"

                  }
                  then {
                    the IUT sends an OkResponse
                    and the IUT sends a Notify containing
                        ElementState containing
                            state corresponding to normal


                  }
            }
```

TP Id	TP_LIS_SIP_BV_05
Test Objective	IUT responds to SIP Unsubscribe with an 200 OK
Reference	ETSI TS 103 479 [1], clause 6.3.4 EENA Next Generation 112 LTD [i.3], clause 4.10 IETF RFC 5985 [3]
PICS Selection	PICS_SIP_SUBSCRIPTION_ELEMENT_STATE
Initial Conditions	

```
with {
                the IUT receives a SubscribeRequest containing
                    "event" set to "emergency-ElementState"

                and the IUT sends an OkResponse
                and the IUT sends a Notify
            }
```

Expected Behaviour

```
ensure that {
                when {
                    the IUT receives a UnsubscribeRequest containing
                        "event" set to "emergency-ElementState"

                  }
                  then {
                    the IUT sends an OkResponse
                    and the IUT sends a Notify
                  }
            }
```

TP Id	TP_LIS_SIP_BV_06
Test Objective	IUT responds to SIP Subscribe for SecurityPosture with an 200 OK
Reference	ETSI TS 103 479 [1], clause 6.3.3 EENA Next Generation 112 LTD [i.3], clause 4.10 IETF RFC 5985 [3]
PICS Selection	PICS_SIP_SUBSCRIPTION_SECURITY_POSTURE
Expected Behaviour	

```
ensure that {
                when {
                    the IUT receives a SubscribeRequest containing
                        "event" set to "emergency-SecurityPosture"

                  }
                  then {
                    the IUT sends an OkResponse
                    and the IUT sends a Notify containing
                        SecurityPosture containing
                            posture corresponding to green


                  }
            }
```

TP Id	TP_LIS_SIP_BV_07
Test Objective	IUT responds to SIP Unsubscribe with an 200 OK
Reference	ETSI TS 103 479 [1], clause 6.3.3 EENA Next Generation 112 LTD [i.3], clause 4.10 IETF RFC 5985 [3]
PICS Selection	PICS_SIP_SUBSCRIPTION_SECURITY_POSTURE
Initial Conditions	

```
with {
                the IUT receives a SubscribeRequest containing
                    "event" set to "emergency-SecurityPosture"

                and the IUT sends an OkResponse
                and the IUT sends a Notify
            }
```

Expected Behaviour

```
ensure that {
                when {
                    the IUT receives a UnsubscribeRequest containing
                        "event" set to "emergency-SecurityPosture"

                  }
                  then {
                    the IUT sends an OkResponse
                    and the IUT sends a Notify
                  }
            }
```

TP Id	TP_LIS_SIP_BV_08
Test Objective	IUT responds to SIP Subscribe for ServiceState with an 200 OK
Reference	ETSI TS 103 479 [1], clause 6.3.5 EENA Next Generation 112 LTD [i.3], clause 4.10 IETF RFC 5985 [3]
PICS Selection	PICS_SIP_SUBSCRIPTION_SERVICE_STATE
Expected Behaviour	

```
ensure that {
                when {
                    the IUT receives a SubscribeRequest containing
                        "event" set to "emergency-ServiceState"

                  }
                  then {
                    the IUT sends an OkResponse
                    and the IUT sends a Notify containing
                        ServiceState containing
                            state corresponding to normal


                  }
            }
```

TP Id	TP_LIS_SIP_BV_09
Test Objective	IUT responds to SIP Unsubscribe with an 200 OK
Reference	ETSI TS 103 479 [1], clause 6.3.5 EENA Next Generation 112 LTD [i.3], clause 4.10 IETF RFC 5985 [3]
PICS Selection	PICS_SIP_SUBSCRIPTION_SERVICE_STATE
Initial Conditions	

```
with {
                the IUT receives a SubscribeRequest containing
                    "event" set to "emergency-ServiceState"

                and the IUT sends an OkResponse
                and the IUT sends a Notify
            }
```

Expected Behaviour

```
ensure that {
                when {
                    the IUT receives a UnsubscribeRequest containing
                        "event" set to "emergency-ServiceState"

                  }
                  then {
                    the IUT sends an OkResponse
                    and the IUT sends a Notify
                  }
            }
```

7.2.1.4 Advanced Mobile Location

TP Id	TP_LIS_AML_BV_01
Test Objective	IUT successfully accepts AML HTTP Data
Reference	ETSI TS 103 479 [1], clause 5.5 EENA Next Generation 112 LTD [i.3], clause 4.10 IETF RFC 5985 [3]
PICS Selection	PICS_LIS_AML_HTTP
Expected Behaviour	

```
ensure that {
                when {
                    the IUT receives an AML_HTTP_Message
                  }
                  then {
                    the IUT sends an AML_HTTP_MessageResponse
                  }
            }
```

TP Id	TP_LIS_AML_BV_02
Test Objective	IUT successfully provides received AML HTTP Data via HELD
Reference	ETSI TS 103 479 [1], clause 5.5 EENA Next Generation 112 LTD [i.3], clause 4.10 IETF RFC 5985 [3]
PICS Selection	PICS_LIS_AML_HTTP and PICS_LIS_GEOMETRY_CIRCLE
Expected Behaviour	

```
ensure that {
                when {
                    the IUT receives an AML_HTTP_Message containing
                        device_number     corresponding to PX_AML_DEVICE_NUMBER,
                        location         corresponding to PX_CIRCLE

                    and the IUT sends an AML_HTTP_MessageResponse
                    and the IUT receives an LocationRequest containing
                        uri        corresponding to PX_AML_DEVICE_NUMBER

                  }
                  then {
                    the IUT sends a LocationResponse containing
                        presence containing
                            "location-info"    corresponding to PX_CIRCLE


                  }
            }
```

TP Id	TP_LIS_AML_BV_03
Test Objective	IUT successfully provides received AML SMS Text via HELD
Reference	ETSI TS 103 479 [1], clause 5.5 EENA Next Generation 112 LTD [i.3], clause 4.10 IETF RFC 5985 [3]
PICS Selection	PICS_LIS_AML_SMS_TEXT and PICS_LIS_GEOMETRY_CIRCLE
Expected Behaviour	

```
ensure that {
                when {
                    the IUT receives an AML_SMS_Text_Message containing
                        device_number     corresponding to PX_AML_DEVICE_NUMBER,
                        location         corresponding to PX_CIRCLE

                    and the IUT receives an LocationRequest containing
                        uri        corresponding to PX_AML_DEVICE_NUMBER

                  }
                  then {
                    the IUT sends a LocationResponse containing
                        presence containing
                            "location-info"    corresponding to PX_CIRCLE


                  }
            }
```

TP Id	TP_LIS_AML_BV_04
Test Objective	IUT successfully provides received AML SMS Data via HELD
Reference	ETSI TS 103 479 [1], clause 5.5 EENA Next Generation 112 LTD [i.3], clause 4.10 IETF RFC 5985 [3]
PICS Selection	PICS_LIS_AML_SMS_DATA and PICS_LIS_GEOMETRY_CIRCLE
Expected Behaviour	

```
ensure that {
                when {
                    the IUT receives an AML_SMS_Data_Message containing
                        device_number     corresponding to PX_AML_DEVICE_NUMBER,
                        location         corresponding to PX_CIRCLE

                    and the IUT receives an LocationRequest containing
                        uri        corresponding to PX_AML_DEVICE_NUMBER

                  }
                  then {
                    the IUT sends a LocationResponse containing
                        presence containing
                            "location-info"    corresponding to PX_CIRCLE


                  }
            }
```

7.2.1.5 Security

TP Id	TP_LIS_TLS_BI_01
Test Objective	IUT rejects HELD requests without a certificate
Reference	ETSI TS 103 479 [1], clause 5.5 EENA Next Generation 112 LTD [i.3], clause 4.10 IETF RFC 5985 [3]
PICS Selection	PICS_LIS_TLS_AUTHENTICATION
Expected Behaviour	

```
ensure that {
                when {
                    the IUT receives any LocationRequest containing
                        not certificate

                  }
                  then {
                    the IUT rejects the Connection
                  }
            }
```

TP Id	TP_LIS_TLS_BI_02
Test Objective	IUT rejects SIP requests without a certificate
Reference	ETSI TS 103 479 [1], clause 5.5 EENA Next Generation 112 LTD [i.3], clause 4.10 IETF RFC 5985 [3]
PICS Selection	PICS_LIS_TLS_AUTHENTICATION
Expected Behaviour	

```
ensure that {
                when {
                    the IUT receives any SubscribeRequest containing
                        not certificate

                  }
                  then {
                    the IUT rejects the Connection
                  }
            }
```

7.2.2 ESRP

7.2.2.1 FindService

TP Id	TP_ESRP_FIND_SERVICE_BV_01
Test Objective	IUT fetches service URN from ECRF and forwards an incoming call with Geodetic Location
Reference	ETSI TS 103 479 [1], clause 5.2.2 EENA Next Generation 112 LTD [i.3], clause 4.4 IETF RFC 5222 [5]
PICS Selection	PICS_ESRP_LOST
Initial Conditions	

```
with {
                    the IUT isConfiguredWith an EcrfConfig containing
                        uri set to PX_ECRF

                    and
                    the ECRF reachable_by a uri PX_ECRF
                    and
                    the DOWNSTREAM reachable_by a uri URI_POLICE_G1
                }
```

Expected Behaviour

```
ensure that {
                when {
                    the IUT receives
                        a request SipInvite containing
                            location corresponding to PX_CIRCLE_IN_G1,
                            service set to "urn:service:sos.police",
                            caller corresponding to PX_CALLER

                }
                then {
                    the IUT sends a request FindServiceRequest containing
                        service     corresponding to POLICE,
                        location corresponding to PX_CIRCLE_IN_G1

                      to the ECRF entity
                    and
                    the ECRF sends a response FindServiceResponse containing
                        service    corresponding to POLICE,
                        uri        corresponding to URI_POLICE_G1

                    and
                    the IUT forwards a request SipInvite containing
                            location corresponding to PX_CIRCLE_IN_G1,
                            service set to "urn:service:sos.police",
                            caller corresponding to PX_CALLER

                    to the DOWNSTREAM entity
                }
            }
```

TP Id	TP_ESRP_FIND_SERVICE_BV_02
Test Objective	IUT fetches service URN from ECRF with redirect and forwards an incoming call with Geodetic Location
Reference	ETSI TS 103 479 [1], clause 5.2.2 EENA Next Generation 112 LTD [i.3], clause 4.4 IETF RFC 5222 [5]
PICS Selection	PICS_ESRP_LOST and PICS_ESRP_LOST_REDIRECT
Initial Conditions	

```
with {
                    the IUT isConfiguredWith an EcrfConfig containing
                        uri set to PX_ECRF

                    and
                    the ECRF reachable_by a uri PX_ECRF
                    and
                    the ECRF_2 reachable_by a uri PX_REDIRECT_TARGET
                    and
                    the DOWNSTREAM reachable_by a uri URI_POLICE_G1
                }
```

Expected Behaviour

ensure that {
when {
the IUT receives
a request SipInvite containing
location corresponding to PX_CIRCLE_IN_G1,
service set to "urn:service:sos.police",
caller corresponding to PX_CALLER
 	
}
then {
the IUT sends a request FindServiceRequest containing
service corresponding to POLICE,
location corresponding to PX_CIRCLE_IN_G1
 	
to the ECRF entity
and
the ECRF sends a response RedirectResponse containing
target corresponding to PX_REDIRECT_TARGET
 	
and
the IUT sends a request FindServiceRequest containing
service corresponding to POLICE,
location corresponding to PX_CIRCLE_IN_G1
 	
to the ECRF_2 entity
and
the ECRF_2 sends a response FindServiceResponse containing
service corresponding to POLICE,
uri corresponding to URI_POLICE_G1
 	
and
the IUT forwards a request SipInvite containing
location corresponding to PX_CIRCLE_IN_G1,
service set to "urn:service:sos.police",
caller corresponding to PX_CALLER
 	
to the DOWNSTREAM entity
}
}

TP Id	TP_ESRP_FIND_SERVICE_BV_03
Test Objective	IUT gets location from LIS and fetches service URN from ECRF and forwards an incoming call without Location
Reference	ETSI TS 103 479 [1], clause 5.2.2 EENA Next Generation 112 LTD [i.3], clause 4.4 IETF RFC 5222 [5]
PICS Selection	PICS_ESRP_HELD and PICS_ESRP_LOST
Initial Conditions	

with {
the IUT isConfiguredWith an EcrfConfig containing
uri set to PX_ECRF
 	
and
the IUT isConfiguredWith a LisConfig containing
uri set to PX_LIS
 	
and
the ECRF reachable_by a uri PX_ECRF
and
the DOWNSTREAM reachable_by a uri URI_POLICE_G1
}	

Expected Behaviour

ensure that {
when {
the IUT receives
a request SipInvite containing
service set to "urn:service:sos.police",
caller corresponding to PX_CALLER
 	
}
then {
the IUT sends a LocationRequest containing
uri corresponding to PX_CALLER
 	
to the LIS entity
and
the IUT sends a LocationResponse containing
presence containing
"location-info" carrying PX_CIRCLE_IN_G1
 	
and
the IUT sends a request FindServiceRequest containing
service corresponding to POLICE,
location corresponding to PX_CIRCLE_IN_G1
 	
to the ECRF entity
and
the ECRF sends a response FindServiceResponse containing
service corresponding to POLICE,
uri corresponding to URI_POLICE_G1
 	
and
the IUT forwards a request SipInvite containing
service set to "urn:service:sos.police",
caller corresponding to PX_CALLER
 	
to the DOWNSTREAM entity
}
}

TP Id	TP_ESRP_FIND_SERVICE_BV_04
Test Objective	IUT resolves location reference and fetches service URN from ECRF and forwards an incoming call with location reference
Reference	ETSI TS 103 479 [1], clause 5.2.2 EENA Next Generation 112 LTD [i.3], clause 4.4 IETF RFC 5222 [5]
PICS Selection	PICS_ESRP_HELD_LOCATION_REFERENCE and PICS_ESRP_LOST
Initial Conditions	

with {
the IUT isConfiguredWith an EcrfConfig containing
uri set to PX_ECRF
 	
and
the ECRF reachable_by a uri PX_ECRF
and
the LIS reachable_by a uri PX_LOCATION_REFERENCE
and
the DOWNSTREAM reachable_by a uri URI_POLICE_G1
}	

Expected Behaviour

ensure that {
when {
the IUT receives a request SipInvite containing
service set to "urn:service:sos.police",
caller corresponding to PX_CALLER,
locationReference corresponding to PX_LOCATION_REFERENCE
 	
}
then {
the IUT sends a LocationDerefenceRequest
to the LIS entity
and
the IUT sends a LocationResponse containing
presence containing
"location-info" carrying PX_CIRCLE_IN_G1
 	
and
the IUT sends a request FindServiceRequest containing
service corresponding to POLICE,
location corresponding to PX_CIRCLE_IN_G1
 	
to the ECRF entity
and
the ECRF sends a response FindServiceResponse containing
service corresponding to POLICE,
uri corresponding to URI_POLICE_G1
 	
and
the IUT forwards a request SipInvite containing
service set to "urn:service:sos.police",
caller corresponding to PX_CALLER
 	
to the DOWNSTREAM entity
}
}

7.2.2.2 Fixed Target

TP Id	TP_ESRP_FIXED_TARGET_SIP_INVITE_BV_01
Test Objective	IUT forwards an incoming call to a fixed target
Reference	ETSI TS 103 479 [1], clause 5.2.2 EENA Next Generation 112 LTD [i.3], clause 4.4 IETF RFC 5222 [5]
PICS Selection	PICS_ESRP_FIXED_TARGET_SUPPORT
Initial Conditions	

with {
the IUT is configured for the DOWNSTREAM entity
}	

Expected Behaviour	

ensure that {
when {
the IUT receives a request SipInvite
}
then {
the IUT forwards thetyped request SipInvite
to the DOWNSTREAM entity
}
}	

7.2.2.3 SIP REGISTER Target

TP Id	TP_ESRP_SIP_REGISTER_TARGET_SIP_REGISTER_BV_01
Test Objective	IUT accepts SIP register requests
Reference	ETSI TS 103 479 [1], clause 5.2.2 EENA Next Generation 112 LTD [i.3], clause 4.4 IETF RFC 5222 [5]
PICS Selection	PICS_ESRP_TARGET_SIP_REGISTER_SUPPORT
Expected Behaviour	

```
ensure that {
                when {
                    the IUT receives
                        a SIP_REGISTER
                        from the DOWNSTREAM entity
                }
                then {
                    the IUT returns
                        a SIP_RESPONSE containing
                            STATUS_CODE set to 200

                        to the DOWNSTREAM entity
                }
            }
```

7.2.2.4 'Dequeue Registration Target

TP Id	TP_ESRP_HTTP_DEQUEUE_REGISTRATION_TARGET_REGISTRATION_BV_01
Test Objective	IUT accepts Dequeue Registration
Reference	ETSI TS 103 479 [1], clause 5.2.2 EENA Next Generation 112 LTD [i.3], clause 4.4 IETF RFC 5222 [5]
PICS Selection	PICS_ESRP_TARGET_DEQUEUE_REGISTRATION_SUPPORT
Expected Behaviour	

```
ensure that {
                when {
                    the IUT receives
                        a DEQUEUE_REGISTRATION_REQUEST
                        from the DOWNSTREAM entity
                }
                then {
                    the IUT returns
                        a DEQUEUE_REGISTRATION_RESPONSE
                    to the DOWNSTREAM entity
                }
            }
```

TP Id	TP_ESRP_HTTP_DEQUEUE_REGISTRATION_TARGET_REGISTRATION_BV_02
Test Objective	IUT sends correct error response for invalid Dequeue Registration
Reference	ETSI TS 103 479 [1], clause 5.2.2 EENA Next Generation 112 LTD [i.3], clause 4.4 IETF RFC 5222 [5]
PICS Selection	PICS_ESRP_TARGET_DEQUEUE_REGISTRATION_SUPPORT
Expected Behaviour	

```
ensure that {
                when {
                    the IUT receives
                        a DEQUEUE_REGISTRATION_REQUEST
                        containing
                            an invalid queue
                        from the DOWNSTREAM entity
                }
                then {
                    the IUT returns
                        a DEQUEUE_REGISTRATION_RESPONSE
                        containing
                            error_response
                    to the DOWNSTREAM entity
                }
            }
```

TP Id	TP_ESRP_HTTP_DEQUEUE_REGISTRATION_TARGET_DEREGISTRATION_BV_01
Test Objective	IUT accepts Dequeue Deregistration
Reference	ETSI TS 103 479 [1], clause 5.2.2 EENA Next Generation 112 LTD [i.3], clause 4.4 IETF RFC 5222 [5]
PICS Selection	PICS_ESRP_TARGET_DEQUEUE_REGISTRATION_SUPPORT
Expected Behaviour	

```
ensure that {
                when {
                    the IUT receives
                        a DEQUEUE_DEREGISTRATION_REQUEST
                        from the DOWNSTREAM entity
                }
                then {
                    the IUT returns
                        a DEQUEUE_DEREGISTRATION_RESPONSE
                    to the DOWNSTREAM entity
                }
            }
```

7.2.2.5 QueueState

TP Id	TP_ESRP_QUEUE_STATE_BV_01
Test Objective	IUT subscribes for Queue State event
Reference	ETSI TS 103 479 [1], clause 5.2.2 EENA Next Generation 112 LTD [i.3], clause 4.4 IETF RFC 5222 [5]
PICS Selection	PICS_ESRP_QUEUE_STATE_SUPPORT
Expected Behaviour	

```
ensure that {
                when {
                    the IUT receives
                        a DEQUEUE_REGISTRATION_REQUEST
                        from the DOWNSTREAM entity
                }
                then {
                    the IUT sends
                        a QUEUE_STATE_SUBSCRIBE_REQUEST
                    to the DOWNSTREAM entity
                }
            }
```

TP Id	TP_ESRP_QUEUE_STATE_BV_02
Test Objective	IUT response to QueueState Notify
Reference	ETSI TS 103 479 [1], clause 5.2.2 EENA Next Generation 112 LTD [i.3], clause 4.4 IETF RFC 5222 [5]
PICS Selection	PICS_ESRP_QUEUE_STATE_SUPPORT
Initial Conditions	

```
with {
                the DOWNSTREAM dequeueRegistered to the IUT entity
            }
```

Expected Behaviour

```
ensure that {
                when {
                    the IUT receives
                        a QUEUE_STATE_NOTIFY_REQUEST
                        from the DOWNSTREAM entity
                }
                then {
                    the IUT returns
                        a QUEUE_STATE_NOTIFY_RESPONSE
                    to the DOWNSTREAM entity
                }
            }
```

TP Id	TP_ESRP_QUEUE_STATE_BV_03
Test Objective	IUT forwards call to active queue
Reference	ETSI TS 103 479 [1], clause 5.2.2 EENA Next Generation 112 LTD [i.3], clause 4.4 IETF RFC 5222 [5]
PICS Selection	PICS_ESRP_QUEUE_STATE_SUPPORT
Initial Conditions	

with {
the DOWNSTREAM dequeueRegistered to the IUT entity
and
the IUT receives
a QUEUE_STATE_NOTIFY_REQUEST containing
queueState set to "active"
 	
from the DOWNSTREAM entity
}	

Expected Behaviour	

ensure that {
when {
the IUT receives a request SipInvite
}
then {
the IUT forwards
the request SipInvite
to the DOWNSTREAM entity
}
}	

TP Id	TP_ESRP_QUEUE_STATE_BV_04
Test Objective	IUT does not forward call to inactive queue
Reference	ETSI TS 103 479 [1], clause 5.2.2 EENA Next Generation 112 LTD [i.3], clause 4.4 IETF RFC 5222 [5]
PICS Selection	PICS_ESRP_QUEUE_STATE_SUPPORT
Initial Conditions	

with {
the IUT default_routes to the DIVERSION entity
and
the DOWNSTREAM dequeueRegistered to the IUT entity
}	

Expected Behaviour	

ensure that {
when {
the IUT receives
a QUEUE_STATE_NOTIFY_REQUEST containing
queueState set to "inactive"
 	
from the DOWNSTREAM entity
and
the IUT receives a request SipInvite
}
then {
the IUT forwards
the request SipInvite
to the DIVERSION entity
}
}	

TP Id	TP_ESRP_QUEUE_STATE_BV_05
Test Objective	IUT does not forward call to disabled queue
Reference	ETSI TS 103 479 [1], clause 5.2.2 EENA Next Generation 112 LTD [i.3], clause 4.4 IETF RFC 5222 [5]
PICS Selection	PICS_ESRP_QUEUE_STATE_SUPPORT
Initial Conditions	

with {
the IUT default_routes to the DIVERSION entity
and
the DOWNSTREAM dequeueRegistered to the IUT entity
}	

Expected Behaviour	

ensure that {
when {
the IUT receives
a QUEUE_STATE_NOTIFY_REQUEST containing
queueState set to "disabled"
 	
from the DOWNSTREAM entity
and
the IUT receives a request SipInvite
}
then {
the IUT forwards
the request SipInvite
to the DIVERSION entity
}
}	

TP Id	TP_ESRP_QUEUE_STATE_BV_06
Test Objective	IUT does not forward call to full queue
Reference	ETSI TS 103 479 [1], clause 5.2.2 EENA Next Generation 112 LTD [i.3], clause 4.4 IETF RFC 5222 [5]
PICS Selection	PICS_ESRP_QUEUE_STATE_SUPPORT
Initial Conditions	

with {
the IUT default_routes to the DIVERSION entity
and
the DOWNSTREAM dequeueRegistered to the IUT entity
}	

Expected Behaviour	

ensure that {
when {
the IUT receives
a QUEUE_STATE_NOTIFY_REQUEST containing
queueState set to "full"
 	
from the DOWNSTREAM entity
and
the IUT receives a request SipInvite
}
then {
the IUT forwards
the request SipInvite
to the DIVERSION entity
}
}	

TP Id	TP_ESRP_QUEUE_STATE_BV_07
Test Objective	IUT forwards call to standby queue
Reference	ETSI TS 103 479 [1], clause 5.2.2 EENA Next Generation 112 LTD [i.3], clause 4.4 IETF RFC 5222 [5]
PICS Selection	PICS_ESRP_QUEUE_STATE_SUPPORT
Initial Conditions	

with {
the DOWNSTREAM dequeueRegistered to the IUT entity
}	

Expected Behaviour	

ensure that {
when {
the IUT receives
a QUEUE_STATE_NOTIFY_REQUEST containing
queueState set to "standby"
 	
from the DOWNSTREAM entity
and
the IUT receives a request SipInvite
}
then {
the IUT forwards
the request SipInvite
to the DOWNSTREAM entity
}
}	

7.2.2.6 Subscriptions/Notifications

TP Id	TP_ESRP_SIP_BV_01
Test Objective	IUT responds to SIP Subscribe for Element State with an 200 OK
Reference	ETSI TS 103 479 [1], clause 5.3 EENA Next Generation 112 LTD [i.3], clause 4.10 IETF RFC 5985 [3]
PICS Selection	PICS_SIP_SUBSCRIPTION_ELEMENT_STATE
Expected Behaviour	

ensure that {
when {
the IUT receives a SubscribeRequest containing
"event" set to "emergency-ElementState"
 	
}
then {
the IUT sends an OkResponse
and the IUT sends a Notify containing
ElementState containing
state corresponding to normal
 	
}
}	

TP Id	TP_ESRP_SIP_BV_02
Test Objective	IUT responds to SIP Unsubscribe with an 200 OK
Reference	ETSI TS 103 479 [1], clause 6.5.3.2 EENA Next Generation 112 LTD [i.3], clause 4.10 IETF RFC 5985 [3]
PICS Selection	PICS_SIP_SUBSCRIPTION_ELEMENT_STATE
Initial Conditions	

with {
the IUT receives a SubscribeRequest containing
"event" set to "emergency-ElementState"
 	
and the IUT sends an OkResponse
and the IUT sends a Notify
}	

Expected Behaviour	

ensure that {
when {
the IUT receives a UnsubscribeRequest containing
"event" set to "emergency-ElementState"
 	
}
then {
the IUT sends an OkResponse
and the IUT sends a Notify
}
}	

TP Id	TP_ESRP_SIP_BV_03
Test Objective	IUT responds to SIP Subscribe for SecurityPosture with an 200 OK
Reference	ETSI TS 103 479 [1], clause 5.3 EENA Next Generation 112 LTD [i.3], clause 4.10 IETF RFC 5985 [3]
PICS Selection	PICS_SIP_SUBSCRIPTION_SECURITY_POSTURE
Expected Behaviour	

ensure that {
when {
the IUT receives a SubscribeRequest containing
"event" set to "emergency-SecurityPosture"
 	
}
then {
the IUT sends an OkResponse
and the IUT sends a Notify containing
SecurityPosture containing
posture corresponding to green
 	
}
}	

TP Id	TP_ESRP_SIP_BV_04
Test Objective	IUT responds to SIP Unsubscribe with an 200 OK
Reference	ETSI TS 103 479 [1], clause 6.5.3.2 EENA Next Generation 112 LTD [i.3], clause 4.10 IETF RFC 5985 [3]
PICS Selection	PICS_SIP_SUBSCRIPTION_SECURITY_POSTURE
Initial Conditions	

with {
the IUT receives a SubscribeRequest containing
"event" set to "emergency-SecurityPosture"
 	
and the IUT sends an OkResponse
and the IUT sends a Notify
}	

Expected Behaviour	

ensure that {
when {
the IUT receives a UnsubscribeRequest containing
"event" set to "emergency-SecurityPosture"
 	
}
then {
the IUT sends an OkResponse
and the IUT sends a Notify
}
}	

TP Id	TP_ESRP_SIP_BV_05
Test Objective	IUT responds to SIP Subscribe for ServiceState with an 200 OK
Reference	ETSI TS 103 479 [1], clause 5.3 EENA Next Generation 112 LTD [i.3], clause 4.10 IETF RFC 5985 [3]
PICS Selection	PICS_SIP_SUBSCRIPTION_SERVICE_STATE
Expected Behaviour	

ensure that {
when {
the IUT receives a SubscribeRequest containing
"event" set to "emergency-ServiceState"
 	
}
then {
the IUT sends an OkResponse
and the IUT sends a Notify containing
ServiceState containing
state corresponding to normal
 	
}
}	

TP Id	TP_ESRP_SIP_BV_06
Test Objective	IUT responds to SIP Unsubscribe with an 200 OK
Reference	ETSI TS 103 479 [1], clause 6.5.3.2 EENA Next Generation 112 LTD [i.3], clause 4.10 IETF RFC 5985 [3]
PICS Selection	PICS_SIP_SUBSCRIPTION_SERVICE_STATE
Initial Conditions	

with {
the IUT receives a SubscribeRequest containing
"event" set to "emergency-ServiceState"
 	
and the IUT sends an OkResponse
and the IUT sends a Notify
}	

Expected Behaviour	

ensure that {
when {
the IUT receives a UnsubscribeRequest containing
"event" set to "emergency-ServiceState"
 	
}
then {
the IUT sends an OkResponse
and the IUT sends a Notify
}
}	

7.2.2.7 Security

TP Id	TP_ESRP_TLS_SIP_INVITE_BI_01
Test Objective	IUT rejects SIP requests without a certificate
Reference	ETSI TS 103 479 [1], clause 6.1.1 EENA Next Generation 112 LTD [i.3], clause 4.10 IETF RFC 5985 [3]
PICS Selection	PICS_ESRP_TLS_AUTHENTICATION
Expected Behaviour	

ensure that {
when {
the IUT receives a request SipInvite containing
not certificate
 	
}
then {
the IUT rejects the Connection
}
}	

7.2.2.8 Headers

TP Id	TP_ESRP_HEADERS_SIP_INVITE_BV_01
Test Objective	IUT adds CallIdentifier header and forwards an incoming call to a fixed target
Reference	ETSI TS 103 479 [1], clause 6.1.2.7
PICS Selection	PICS_ESRP_FIXED_TARGET_SUPPORT
Initial Conditions	

with {
the IUT is configured for the DOWNSTREAM entity
}	

Expected Behaviour	

ensure that {
when {
the IUT receives a request SipInvite containing
not callIdHeader
 	
}
then {
the IUT forwards
a request SipInvite containing
callIdHeader
 	
to the DOWNSTREAM entity
}
}	

TP Id	TP_ESRP_HEADERS_SIP_INVITE_BV_02
Test Objective	IUT adds IncidentTrackingIdentifier header and forwards an incoming call to a fixed target
Reference	ETSI TS 103 479 [1], clause 6.1.2.7
PICS Selection	PICS_ESRP_FIXED_TARGET_SUPPORT
Initial Conditions	

with {
the IUT is configured for the DOWNSTREAM entity
}	

Expected Behaviour	

ensure that {
when {
the IUT receives a request SipInvite containing
not incidentIdHeader
 	
}
then {
the IUT forwards
a request SipInvite containing
incidentIdHeader
 	
to the DOWNSTREAM entity
}
}	

7.2.3 ECRF

7.2.3.1 LoST

TP Id	TP_ECRF_LOST_BV_01
Test Objective	IUT successfully responds with a service URI for a Point in the service boundary
Reference	ETSI TS 103 479 [1], clause 6.4.1 EENA Next Generation 112 LTD [i.3], clause 4.4 IETF RFC 5222 [5]
PICS Selection	PICS_ECRF_FIND_SERVICE and PICS_ECRF_GEOMETRY_POINT
Initial Conditions	

with {
the IUT has a mapping PX_POLICE_G1
}	

Expected Behaviour	

ensure that {
when {
the IUT receives a request FindServiceRequest containing
service corresponding to POLICE,
location corresponding to PX_POINT_IN_G1
 	
}
then {
the IUT sends a response FindServiceResponse containing
service corresponding to POLICE,
uri corresponding to URI_POLICE_G1
 	
}
}	

TP Id	TP_ECRF_LOST_BV_02
Test Objective	IUT successfully responds with a service URI for a Circle in the service boundary
Reference	ETSI TS 103 479 [1], clause 6.4.1 EENA Next Generation 112 LTD [i.3], clause 4.4 IETF RFC 5222 [5]
PICS Selection	PICS_ECRF_FIND_SERVICE and PICS_ECRF_GEOMETRY_CIRCLE
Initial Conditions	

with {
the IUT has a mapping PX_POLICE_G1
}	

Expected Behaviour	

ensure that {
when {
the IUT receives a request FindServiceRequest containing
service corresponding to POLICE,
location corresponding to PX_CIRCLE_IN_G1
 	
}
then {
the IUT sends a response FindServiceResponse containing
service corresponding to POLICE,
uri corresponding to URI_POLICE_G1
 	
}
}	

TP Id	TP_ECRF_LOST_BV_03
Test Objective	IUT successfully responds with an error response for an unknown Service URN in the service boundary
Reference	ETSI TS 103 479 [1], clause 6.4.1 EENA Next Generation 112 LTD [i.3], clause 4.4 IETF RFC 5222 [5]
PICS Selection	PICS_ECRF_FIND_SERVICE
Initial Conditions	

with {
the IUT has no Mapping containing
service set to "urn:some:unknown:service"
 	
}	

Expected Behaviour	

ensure that {
when {
the IUT receives a request FindServiceRequest containing
service set to "urn:some:unknown:service"
 	
}
then {
the IUT sends a response ErrorResponse containing
errors indicating value "serviceNotImplemented"
 	
}
}	

TP Id	TP_ECRF_LOST_BV_04
Test Objective	IUT successfully responds with an error response for an unrecognized location profile
Reference	ETSI TS 103 479 [1], clause 6.4.1 EENA Next Generation 112 LTD [i.3], clause 4.4 IETF RFC 5222 [5]
PICS Selection	PICS_ECRF_FIND_SERVICE
Initial Conditions	

with {
the IUT has no Mapping containing
service set to "urn:some:unknown:service"
 	
}	

Expected Behaviour	

ensure that {
when {
the IUT receives a request FindServiceRequest containing
profile set to "someUnknownProfile"
 	
}
then {
the IUT sends a response ErrorResponse containing
errors indicating value "locationProfileUnrecognized"
 	
}
}	

TP Id	TP_ECRF_LOST_BV_05
Test Objective	IUT successfully includes the service boundary by value
Reference	ETSI TS 103 479 [1], clause 6.4.1 EENA Next Generation 112 LTD [i.3], clause 4.4 IETF RFC 5222 [5]
PICS Selection	PICS_ECRF_FIND_SERVICE and PICS_ECRF_GEOMETRY_POINT
Initial Conditions	

with {
the IUT has a mapping PX_POLICE_G1
}	

Expected Behaviour	

ensure that {
when {
the IUT receives a request FindServiceRequest containing
service corresponding to POLICE,
location corresponding to PX_POINT_IN_G1,
serviceBoundary set to "value"
 	
}
then {
the IUT sends a response FindServiceResponse containing
service corresponding to POLICE,
uri corresponding to URI_POLICE_G1,
geometry corresponding to PX_G1
 	
}
}	

TP Id	TP_ECRF_LOST_BV_06
Test Objective	IUT successfully responds with the service URI for intersecting Circle
Reference	ETSI TS 103 479 [1], clause 6.4.1 EENA Next Generation 112 LTD [i.3], clause 4.4 IETF RFC 5222 [5]
PICS Selection	PICS_ECRF_FIND_SERVICE and PICS_ECRF_GEOMETRY_CIRCLE
Initial Conditions	

with {
the IUT has a mapping PX_POLICE_G1
}	

Expected Behaviour	

ensure that {
when {
the IUT receives a request FindServiceRequest containing
service corresponding to POLICE,
location corresponding to PX_CIRCLE_INTERSECT_ONLY_G1
 	
}
then {
the IUT sends a response FindServiceResponse containing
service corresponding to POLICE,
uri corresponding to URI_POLICE_G1
 	
}
}	

TP Id	TP_ECRF_LOST_BV_07
Test Objective	IUT successfully responds with the service URI for Circle intersecting multiple service boundaries
Reference	ETSI TS 103 479 [1], clause 6.4.1 EENA Next Generation 112 LTD [i.3], clause 4.4 IETF RFC 5222 [5]
PICS Selection	PICS_ECRF_FIND_SERVICE and PICS_ECRF_GEOMETRY_CIRCLE
Initial Conditions	

with {
the IUT has a mapping PX_POLICE_G1
and the IUT has a mapping PX_POINT_IN_G2
}	

Expected Behaviour	

ensure that {
when {
the IUT receives a request FindServiceRequest containing
service corresponding to POLICE,
location corresponding to PX_CIRCLE_INTERSECT_MAINLY_G1_AND_G2
 	
}
then {
the IUT sends a response FindServiceResponse containing
service corresponding to POLICE,
uri corresponding to URI_POLICE_G1
 	
}
}	

TP Id	TP_ECRF_LOST_BV_08
Test Objective	IUT successfully responds with the service URI for Circle a boundary with multiple services
Reference	ETSI TS 103 479 [1], clause 6.4.1 EENA Next Generation 112 LTD [i.3], clause 4.4 IETF RFC 5222 [5]
PICS Selection	PICS_ECRF_FIND_SERVICE and PICS_ECRF_GEOMETRY_CIRCLE
Initial Conditions	

with {
the IUT has a mapping PX_POLICE_G1
and the IUT has a mapping PX_FIRE_G1
}	

Expected Behaviour	

ensure that {
when {
the IUT receives a request FindServiceRequest containing
service corresponding to FIRE,
location corresponding to PX_CIRCLE_IN_G1
 	
}
then {
the IUT sends a response FindServiceResponse containing
service corresponding to FIRE,
uri corresponding to URI_FIRE_G1
 	
}
}	

TP Id	TP_ECRF_LOST_BV_09
Test Objective	IUT successfully responds with configured service types for a ListServices request
Reference	ETSI TS 103 479 [1], clause 6.4.3 EENA Next Generation 112 LTD [i.3], clause 4.4 IETF RFC 5222 [5]
PICS Selection	PICS_ECRF_LIST_SERVICES
Expected Behaviour	

```
ensure that {
                when {
                    the IUT receives a request ListServicesRequest containing
                        service set to "urn:service:sos"

                }
                then {
                    the IUT sends a response ListServicesResponse containing
                        services     corresponding to PX_CONFIGURED_SERVICES

                }
            }
```

TP Id	TP_ECRF_LOST_BV_10
Test Objective	IUT successfully responds with configured service types for a ListServicesByLocation request
Reference	ETSI TS 103 479 [1], clause 6.4.3 EENA Next Generation 112 LTD [i.3], clause 4.4 IETF RFC 5222 [5]
PICS Selection	PICS_ECRF_LIST_SERVICES_BY_LOCATION
Initial Conditions	

```
with {
                the IUT has a mapping PX_POLICE_G1
                and the IUT has a mapping PX_FIRE_G1
            }
```

Expected Behaviour

```
ensure that {
                when {
                    the IUT receives a request ListServicesByLocationRequest containing
                        location corresponding to PX_POINT_IN_G1,
                        service     set to "urn:service:sos"

                }
                then {
                    the IUT sends a response ListServicesByLocationResponse containing
                        serviceList    carrying POLICE,
                        serviceList carrying FIRE

                }
            }
```

TP Id	TP_ECRF_LOST_BV_11
Test Objective	IUT successfully responds with configured service types for a ListServices request without service element
Reference	ETSI TS 103 479 [1], clause 6.4.3 EENA Next Generation 112 LTD [i.3], clause 4.4 IETF RFC 5222 [5]
PICS Selection	PICS_ECRF_LIST_SERVICES
Expected Behaviour	

```
ensure that {
                when {
                    the IUT receives a request ListServicesRequest
                }
                then {
                    the IUT sends a response ListServicesResponse containing
                        services     corresponding to PX_CONFIGURED_SERVICES

                }
            }
```

TP Id	TP_ECRF_LOST_BV_12
Test Objective	IUT successfully responds with configured service types for a ListServicesByLocation request without service
Reference	ETSI TS 103 479 [1], clause 6.4.3 EENA Next Generation 112 LTD [i.3], clause 4.4 IETF RFC 5222 [5]
PICS Selection	PICS_ECRF_LIST_SERVICES_BY_LOCATION
Initial Conditions	

with {
the IUT has a mapping PX_POLICE_G1
and the IUT has a mapping PX_FIRE_G1
}	

Expected Behaviour	

ensure that {
when {
the IUT receives a request ListServicesByLocationRequest containing
location corresponding to PX_POINT_IN_G1,
not service
 	
}
then {
the IUT sends a response ListServicesByLocationResponse containing
serviceList carrying POLICE,
serviceList carrying FIRE
 	
}
}	

TP Id	TP_ECRF_LOST_BV_13
Test Objective	IUT successfully responds with a service URI for an ellipse in the service boundary
Reference	ETSI TS 103 479 [1], clause 6.4.1 EENA Next Generation 112 LTD [i.3], clause 4.4 IETF RFC 5222 [5]
PICS Selection	PICS_ECRF_FIND_SERVICE and PICS_ECRF_GEOMETRY_ELLIPSE
Initial Conditions	

with {
the IUT has a mapping PX_POLICE_G1
}	

Expected Behaviour	

ensure that {
when {
the IUT receives a request FindServiceRequest containing
service corresponding to POLICE,
location corresponding to PX_ELLIPSE_IN_G1
 	
}
then {
the IUT sends a response FindServiceResponse containing
service corresponding to POLICE,
uri corresponding to URI_POLICE_G1
 	
}
}	

TP Id	TP_ECRF_LOST_BV_14
Test Objective	IUT successfully responds with a service URI for an arcband in the service boundary
Reference	ETSI TS 103 479 [1], clause 6.4.1 EENA Next Generation 112 LTD [i.3], clause 4.4 IETF RFC 5222 [5]
PICS Selection	PICS_ECRF_FIND_SERVICE and PICS_ECRF_GEOMETRY_ARCBAND
Initial Conditions	

with {
the IUT has a mapping PX_POLICE_G1
}	

Expected Behaviour	

ensure that {
when {
the IUT receives a request FindServiceRequest containing
service corresponding to POLICE,
location corresponding to PX_ARCBAND_IN_G1
 	
}
then {
the IUT sends a response FindServiceResponse containing
service corresponding to POLICE,
uri corresponding to URI_POLICE_G1
 	
}
}	

TP Id	TP_ECRF_LOST_BV_15
Test Objective	IUT successfully responds with a service URI for an polygon in the service boundary
Reference	ETSI TS 103 479 [1], clause 6.4.1 EENA Next Generation 112 LTD [i.3], clause 4.4 IETF RFC 5222 [5]
PICS Selection	PICS_ECRF_FIND_SERVICE and PICS_ECRF_GEOMETRY_POLYGON
Initial Conditions	

with {
the IUT has a mapping PX_POLICE_G1
}	

Expected Behaviour	

ensure that {
when {
the IUT receives a request FindServiceRequest containing
service corresponding to POLICE,
location corresponding to PX_POLYGON_IN_G1
 	
}
then {
the IUT sends a response FindServiceResponse containing
service corresponding to POLICE,
uri corresponding to URI_POLICE_G1
 	
}
}	

TP Id	TP_ECRF_LOST_BV_16
Test Objective	IUT responds with redirect response for a findServiceRequest
Reference	ETSI TS 103 479 [1], clause 6.4.1 EENA Next Generation 112 LTD [i.3], clause 4.4 IETF RFC 5222 [5]
PICS Selection	PICS_ECRF_FIND_SERVICE and PICS_ECRF_GEOMETRY_POINT and PICS_ECRF_REDIRECT
Initial Conditions	

with {
the IUT has no typed mapping Mapping containing
geometry set to PX_G3
 	
}	

Expected Behaviour	

ensure that {
when {
the IUT receives a request FindServiceRequest containing
service corresponding to POLICE,
location corresponding to PX_POINT_IN_G3,
recursive set to False
 	
}
then {
the IUT sends a response RedirectResponse containing
target corresponding to PX_REDIRECT_TARGET
 	
}
}	

TP Id	TP_ECRF_LOST_BV_17
Test Objective	IUT responds with redirect response for a listServiceByLocation
Reference	ETSI TS 103 479 [1], clause 6.4.3 EENA Next Generation 112 LTD [i.3], clause 4.4 IETF RFC 5222 [5]
PICS Selection	PICS_ECRF_LIST_SERVICES_BY_LOCATION and PICS_ECRF_GEOMETRY_POINT and PICS_ECRF_REDIRECT
Initial Conditions	

with {
the IUT has no typed mapping Mapping containing
geometry set to PX_G3
 	
}	

Expected Behaviour	

ensure that {
when {
the IUT receives a request ListServicesByLocationRequest containing
location corresponding to PX_POINT_IN_G3,
recursive set to False
 	
}
then {
the IUT sends a response RedirectResponse containing
target corresponding to PX_REDIRECT_TARGET
 	
}
}	

TP Id	TP_ECRF_LOST_BV_18
Test Objective	IUT responds with redirect response for a findServiceRequest with recursive set to false
Reference	ETSI TS 103 479 [1], clause 6.4.3 EENA Next Generation 112 LTD [i.3], clause 4.4 IETF RFC 5222 [5]
PICS Selection	PICS_ECRF_FIND_SERVICE and PICS_ECRF_GEOMETRY_POINT and PICS_ECRF_REDIRECT and PICS_ECRF_RECURSION
Initial Conditions	

with {
the IUT has no typed mapping Mapping containing
geometry set to PX_G3
 	
}	

Expected Behaviour	

ensure that {
when {
the IUT receives a request FindServiceRequest containing
service corresponding to POLICE,
location corresponding to PX_POINT_IN_G3,
recursive set to False
 	
}
then {
the IUT sends a response RedirectResponse containing
target corresponding to PX_REDIRECT_TARGET
 	
}
}	

TP Id	TP_ECRF_LOST_BV_19
Test Objective	IUT responds with redirect response for a listServiceByLocation with recursive set to false
Reference	ETSI TS 103 479 [1], clause 6.4.3 EENA Next Generation 112 LTD [i.3], clause 4.4 IETF RFC 5222 [5]
PICS Selection	PICS_ECRF_LIST_SERVICES_BY_LOCATION and PICS_ECRF_GEOMETRY_POINT and PICS_ECRF_REDIRECT and PICS_ECRF_RECURSION
Initial Conditions	

with {
the IUT has no typed mapping Mapping containing
geometry set to PX_G3
 	
}	

Expected Behaviour	

ensure that {
when {
the IUT receives a request ListServicesByLocationRequest containing
location corresponding to PX_POINT_IN_G3,
recursive set to False
 	
}
then {
the IUT sends a response RedirectResponse containing
target corresponding to PX_REDIRECT_TARGET
 	
}
}	

7.2.3.2 Errors

TP Id	TP_ECRF_LOST_BI_01
Test Objective	IUT responds to a findServiceRequest with an loop error response
Reference	ETSI TS 103 479 [1], clause 6.4.1 EENA Next Generation 112 LTD [i.3], clause 4.4 IETF RFC 5222 [5]
PICS Selection	PICS_ECRF_FIND_SERVICE and PICS_ECRF_LOOP_DETECTION
Expected Behaviour	

```
ensure that {
                when {
                    the IUT receives a request FindServiceRequest containing
                        path carrying PX_IUT_NAPTR

                }
                then {
                    the IUT sends a response ErrorResponse containing
                        errors indicating value "loop"

                }
            }
```

TP Id	TP_ECRF_LOST_BI_02
Test Objective	IUT responds to a listServiceByLocation with an loop error response
Reference	ETSI TS 103 479 [1], clause 6.4.3 EENA Next Generation 112 LTD [i.3], clause 4.4 IETF RFC 5222 [5]
PICS Selection	PICS_ECRF_LIST_SERVICES_BY_LOCATION and PICS_ECRF_LOOP_DETECTION
Expected Behaviour	

```
ensure that {
                when {
                    the IUT receives a request ListServicesByLocationRequest containing
                        path carrying PX_IUT_NAPTR

                }
                then {
                    the IUT sends a response ErrorResponse containing
                        errors indicating value "loop"

                }
            }
```

TP Id	TP_ECRF_LOST_BI_03
Test Objective	IUT responds to an invalid findService with an badRequest error response
Reference	ETSI TS 103 479 [1], clause 6.4.1 EENA Next Generation 112 LTD [i.3], clause 4.4 IETF RFC 5222 [5]
PICS Selection	PICS_ECRF_FIND_SERVICE
Expected Behaviour	

```
ensure that {
                when {
                    the IUT receives a request InvalidRequest
                }
                then {
                    the IUT sends a response ErrorResponse containing
                        errors indicating value "badRequest"

                }
            }
```

7.2.3.3 Subscriptions/Notifications

TP Id	TP_ECRF_SIP_BV_01
Test Objective	IUT responds to SIP Subscribe for Element State with an 200 OK
Reference	ETSI TS 103 479 [1], clause 6.3.4 EENA Next Generation 112 LTD [i.3], clause 4.10 IETF RFC 5985 [3]
PICS Selection	PICS_SIP_SUBSCRIPTION_ELEMENT_STATE
Expected Behaviour	

```
ensure that {
                when {
                    the IUT receives a SubscribeRequest containing
                        "event" set to "emergency-ElementState"

                  }
                  then {
                    the IUT sends an OkResponse
                    and the IUT sends a Notify containing
                        ElementState containing
                            state corresponding to normal


                  }
            }
```

TP Id	TP_ECRF_SIP_BV_02
Test Objective	IUT responds to SIP Unsubscribe with an 200 OK
Reference	ETSI TS 103 479 [1], clause 6.3.4 EENA Next Generation 112 LTD [i.3], clause 4.10 IETF RFC 5985 [3]
PICS Selection	PICS_SIP_SUBSCRIPTION_ELEMENT_STATE
Initial Conditions	

```
with {
                the IUT receives a SubscribeRequest containing
                    "event" set to "emergency-ElementState"

                and the IUT sends an OkResponse
                and the IUT sends a Notify
            }
```

Expected Behaviour

```
ensure that {
                when {
                    the IUT receives a UnsubscribeRequest containing
                        "event" set to "emergency-ElementState"

                  }
                  then {
                    the IUT sends an OkResponse
                    and the IUT sends a Notify
                  }
            }
```

TP Id	TP_ECRF_SIP_BV_03
Test Objective	IUT responds to SIP Subscribe for SecurityPosture with an 200 OK
Reference	ETSI TS 103 479 [1], clause 6.3.3 EENA Next Generation 112 LTD [i.3], clause 4.10 IETF RFC 5985 [3]
PICS Selection	PICS_SIP_SUBSCRIPTION_SECURITY_POSTURE
Expected Behaviour	

```
ensure that {
                when {
                    the IUT receives a SubscribeRequest containing
                        "event" set to "emergency-SecurityPosture"

                  }
                  then {
                    the IUT sends an OkResponse
                    and the IUT sends a Notify containing
                        SecurityPosture containing
                            posture corresponding to green


                  }
            }
```

TP Id	TP_ECRF_SIP_BV_04
Test Objective	IUT responds to SIP Unsubscribe with an 200 OK
Reference	ETSI TS 103 479 [1], clause 6.3.3 EENA Next Generation 112 LTD [i.3], clause 4.10 IETF RFC 5985 [3]
PICS Selection	PICS_SIP_SUBSCRIPTION_SECURITY_POSTURE
Initial Conditions	

```
with {
                the IUT receives a SubscribeRequest containing
                    "event" set to "emergency-SecurityPosture"

                and the IUT sends an OkResponse
                and the IUT sends a Notify
            }
```

Expected Behaviour

```
ensure that {
                when {
                    the IUT receives a UnsubscribeRequest containing
                        "event" set to "emergency-SecurityPosture"

                  }
                  then {
                    the IUT sends an OkResponse
                    and the IUT sends a Notify
                  }
            }
```

TP Id	TP_ECRF_SIP_BV_05
Test Objective	IUT responds to SIP Subscribe for ServiceState with an 200 OK
Reference	ETSI TS 103 479 [1], clause 6.3.5 EENA Next Generation 112 LTD [i.3], clause 4.10 IETF RFC 5985 [3]
PICS Selection	PICS_SIP_SUBSCRIPTION_SERVICE_STATE
Expected Behaviour	

ensure that {
when {
the IUT receives a SubscribeRequest containing
"event" set to "emergency-ServiceState"
 	
}
then {
the IUT sends an OkResponse
and the IUT sends a Notify containing
ServiceState containing
state corresponding to normal
 	
}
}	

TP Id	TP_ECRF_SIP_BV_06
Test Objective	IUT responds to SIP Unsubscribe with an 200 OK
Reference	ETSI TS 103 479 [1], clause 6.3.5 EENA Next Generation 112 LTD [i.3], clause 4.10 IETF RFC 5985 [3]
PICS Selection	PICS_SIP_SUBSCRIPTION_SERVICE_STATE
Initial Conditions	

with {
the IUT receives a SubscribeRequest containing
"event" set to "emergency-ServiceState"
 	
and the IUT sends an OkResponse
and the IUT sends a Notify
}	

Expected Behaviour	

ensure that {
when {
the IUT receives a UnsubscribeRequest containing
"event" set to "emergency-ServiceState"
 	
}
then {
the IUT sends an OkResponse
and the IUT sends a Notify
}
}	

7.2.3.4 Security

TP Id	TP_ECRF_TLS_BI_01
Test Objective	IUT rejects requests without a certificate
Reference	ETSI TS 103 479 [1], clause 6.4.1.4 EENA Next Generation 112 LTD [i.32], clause 4.10 IETF RFC 5985 [3]
PICS Selection	PICS_ECRF_TLS_AUTHENTICATION
Expected Behaviour	

ensure that {
when {
the IUT receives any FindServiceRequest containing
not certificate
 	
}
then {
the IUT rejects the Connection
}
}	

TP Id	TP_ECRF_TLS_BI_02
Test Objective	IUT rejects SIP requests without a certificate
Reference	ETSI TS 103 479 [1], clause 6.4.1.4 EENA Next Generation 112 LTD [i.3], clause 4.10 IETF RFC 5985 [3]
PICS Selection	PICS_ECRF_TLS_AUTHENTICATION
Expected Behaviour	

ensure that {
when {
the IUT receives any SubscribeRequest containing
not certificate
 	
}
then {
the IUT rejects the Connection
}
}	

7.2.4 PSAP

7.2.4.1 Invite

TP Id	TP_PSAP_SIP_INVITE_BV_01
Test Objective	IUT successfully handles SIP INVITE with different service urn and ULAW via UDP, with no SIP registration
Reference	ETSI TS 103 479 [1], clauses 5.4, 6.1.1 and 6.1.2.2 Next Generation 112 LTD [i.3], clause 4.7 IETF RFC 3261 [6] IETF RFC 5301 [7]
Configuration	CFG_PSAP_01
PICS Selection	PICS_PSAP_S_SIP_UDP1 and PICS_PSAP_E_SIP_URN3 and PICS_PSAP_B_SDP_ULA1 and PICS_PSAP_S_SIP_NO_REGISTRATION
Initial Conditions	

with {
the IUT acceptingIncomingCalls
}	

Expected Behaviour	

ensure that {
when {
the IUT receives a SIP_INVITE containing
Request_URI indicating value PX_PSAP_SERVICE_URN,
Content_Type indicating value "application/sdp",
Body containing
SDP_ULAW
 	
}
then {
the IUT establishesIncomingCall
}
}	

TP Id	TP_PSAP_SIP_INVITE_BV_02
Test Objective	IUT successfully handles SIP INVITE with different service urn and ULAW via UDP, with SIP registration
Reference	ETSI TS 103 479 [1], clauses 5.4, 6.1.1 and 6.1.2.2 Next Generation 112 LTD [i.3], clause 4.7 IETF RFC 3261 [6] IETF RFC 5301 [7]
Configuration	CFG_PSAP_01
PICS Selection	PICS_PSAP_S_SIP_UDP1 and PICS_PSAP_E_SIP_URN3 and PICS_PSAP_B_SDP_ULA1 and PICS_PSAP_S_SIP_REGISTRATION
Initial Conditions	

```
with {
the IUT isRegistered to the ESRP and
the IUT acceptingIncomingCalls
            }
```

Expected Behaviour

```
ensure that {
    when {
        the IUT receives a SIP_INVITE containing
            Request_URI indicating value PX_PSAP_SERVICE_URN,
            Content_Type indicating value "application/sdp",
            Body containing
                SDP_ULAW


    }
    then {
        the IUT establishesIncomingCall
    }
}
```

TP Id	TP_PSAP_SIP_INVITE_BV_03
Test Objective	IUT successfully handles SIP INVITE with different service urn and ALAW via UDP, with no SIP registration
Reference	ETSI TS 103 479 [1], clauses 5.4, 6.1.1 and 6.1.2.2 Next Generation 112 LTD [i.3], clause 4.7 IETF RFC 3261 [6] IETF RFC 5301 [7]
Configuration	CFG_PSAP_01
PICS Selection	PICS_PSAP_S_SIP_UDP1 and PICS_PSAP_E_SIP_URN3 and PICS_PSAP_B_SDP_ALA1 and PICS_PSAP_S_SIP_NO_REGISTRATION
Initial Conditions	

```
with {
the IUT acceptingIncomingCalls
            }
```

Expected Behaviour

```
ensure that {
    when {
        the IUT receives a SIP_INVITE containing
            Request_URI indicating value PX_PSAP_SERVICE_URN,
            Content_Type indicating value "application/sdp",
            Body containing
                SDP_ALAW


    }
    then {
        the IUT establishesIncomingCall
    }
}
```

TP Id	TP_PSAP_SIP_INVITE_BV_04
Test Objective	IUT successfully handles SIP INVITE with different service urn and ALAW via UDP, with SIP registration
Reference	ETSI TS 103 479 [1], clauses 5.4, 6.1.1 and 6.1.2.2 Next Generation 112 LTD [i.3], clause 4.7 IETF RFC 3261 [6] IETF RFC 5301 [7]
Configuration	CFG_PSAP_01
PICS Selection	PICS_PSAP_S_SIP_UDP1 and PICS_PSAP_E_SIP_URN3 and PICS_PSAP_B_SDP_ALA1 and PICS_PSAP_S_SIP_REGISTRATION
Initial Conditions	

with {
the IUT isRegistered to the ESRP and
the IUT acceptingIncomingCalls
}	

Expected Behaviour	

ensure that {
when {
the IUT receives a SIP_INVITE containing
Request_URI indicating value PX_PSAP_SERVICE_URN,
Content_Type indicating value "application/sdp",
Body containing
SDP_ALAW
 	
}
then {
the IUT establishesIncomingCall
}
}	

TP Id	TP_PSAP_SIP_INVITE_BV_05
Test Objective	IUT successfully handles SIP INVITE with different service urn via TCP
Reference	ETSI TS 103 479 [1], clauses 5.4, 6.1.1 and 6.1.2.2 Next Generation 112 LTD [i.3], clause 4.7 IETF RFC 3261 [6] IETF RFC 5301 [7]
Configuration	CFG_PSAP_01
PICS Selection	PICS_PSAP_S_SIP_TCP1 and PICS_PSAP_E_SIP_URN3 and PICS_PSAP_B_SDP_ULA1 and PICS_PSAP_S_SIP_NO_REGISTRATION
Initial Conditions	

with {
the IUT acceptingIncomingCalls
}	

Expected Behaviour	

ensure that {
when {
the IUT receives a SIP_INVITE containing
Request_URI indicating value PX_PSAP_SERVICE_URN,
Content_Type indicating value "application/sdp",
Body containing
SDP_ULAW
 	
}
then {
the IUT establishesIncomingCall
}
}	

TP Id	TP_PSAP_SIP_INVITE_BV_06
Test Objective	IUT successfully handles SIP INVITE with SDP and PIDF-LO content
Reference	ETSI TS 103 479 [1], clauses 5.4, 6.1.1 and 6.1.2.2 Next Generation 112 LTD [i.3], clause 4.7 IETF RFC 3261 [6] IETF RFC 5301 [7]
Configuration	CFG_PSAP_01
PICS Selection	PICS_PSAP_S_SIP_TCP1 and PICS_PSAP_E_SIP_URN1 and PICS_PSAP_B_SDP_ULA1 and PICS_PSAP_S_SIP_NO_REGISTRATION
Initial Conditions	

with {
the IUT acceptingIncomingCalls
}	

Expected Behaviour	

ensure that {
when {
the IUT receives a SIP_INVITE containing
Request_URI indicating value PX_PSAP_SERVICE_URN,
Content_Type indicating value "multipart/mixed",
Body containing
SDP_AND_PIDF_MULTIPART
 	
}
then {
the IUT establishesIncomingCall
}
}	

TP Id	TP_PSAP_SIP_INVITE_BV_07
Test Objective	IUT successfully handles SIP INVITE without service URN
Reference	ETSI TS 103 479 [1], clauses 5.4 and 6.1.2.2 Next Generation 112 LTD [i.3], clause 4.7 IETF RFC 3261 [6]
Configuration	CFG_PSAP_01
PICS Selection	PICS_PSAP_S_SIP_UDP1 and PICS_PSAP_A_SIP_BSC1 and PICS_PSAP_B_SDP_ULA1 and PICS_PSAP_S_SIP_NO_REGISTRATION
Initial Conditions	

with {
the IUT acceptingIncomingCalls
}	

Expected Behaviour	

ensure that {
when {
the IUT receives a SIP_INVITE containing
Request_URI indicating value "sip:psap@city.com",
Content_Type indicating value "application/sdp",
Body containing
SDP_ULAW
 	
}
then {
the IUT establishesIncomingCall
}
}	

TP Id	TP_PSAP_SIP_INVITE_BV_08
Test Objective	IUT successfully handles SIP INVITE with Geolocation header
Reference	ETSI TS 103 479 [1], clauses 5.4, 6.1.1 and 6.1.2.2 Next Generation 112 LTD [i.3], clause 4.7 IETF RFC 3261 [6] IETF RFC 5301 [7] IETF RFC 6442 [10]
Configuration	CFG_PSAP_01
PICS Selection	PICS_PSAP_S_SIP_UDP1 and PICS_PSAP_E_SIP_URN3 and PICS_PSAP_B_SDP_ULA1 and PICS_PSAP_S_SIP_REGISTRATION
Initial Conditions	

with {
the IUT isRegistered to the ESRP and
the IUT acceptingIncomingCalls
}	

Expected Behaviour	

ensure that {
when {
the IUT receives a SIP_INVITE containing
Request_URI indicating value PX_PSAP_SERVICE_URN,
Geolocation indicating value PX_GEOLOCATION,
Content_Type indicating value "multipart/mixed",
Body containing
SDP_AND_PIDF_MULTIPART
 	
}
then {
the IUT establishesIncomingCall
}
}	

7.2.4.2 Ack

TP Id	TP_PSAP_SIP_ACK_BV_01
Test Objective	IUT after sending 200 OK INVITE successfully handles an incoming SIP ACK to accept the emergency call
Reference	ETSI TS 103 479 [1], clauses 5.4, 6.1.1 and 6.1.2.2 Next Generation 112 LTD [i.3], clause 4.7 IETF RFC 3261 [6] IETF RFC 5301 [7]
Configuration	CFG_PSAP_01
PICS Selection	PICS_PSAP_S_SIP_UDP1 and PICS_PSAP_E_SIP_URN3 and PICS_PSAP_S_SIP_REGISTRATION
Initial Conditions	

with {
the IUT isRegistered to the ESRP and
the IUT acceptingIncomingCalls
}	

Expected Behaviour	

ensure that {
when {
the IUT sends a SIP_200OK_INVITE
}
then {
the IUT receives a SIP_ACK and
the IUT establishesIncomingCall
}
}	

7.2.4.3 Bye

TP Id	TP_PSAP_SIP_BYE_BV_01
Test Objective	IUT successfully handles an incoming SIP BYE
Reference	ETSI TS 103 479 [1], clauses 5.4 and 6.1.2.2 ETSI TS 103 698 Next Generation 112 LTD [i.3], clause 4.7 IETF RFC 3261 [6] IETF RFC 5301 [7]
Configuration	CFG_PSAP_01
PICS Selection	PICS_PSAP_S_SIP_BYE1 and PICS_PSAP_S_SIP_TCP1 and PICS_PSAP_E_SIP_URN1 and PICS_PSAP_B_SDP_ULA1 and PICS_PSAP_S_SIP_NO_REGISTRATION
Initial Conditions	

with {
the IUT inAnActiveIncomingCall
}
Expected Behaviour	

ensure that {
when {
the IUT receives a SIP_BYE
}
then {
the IUT sends a SIP_200OK_BYE
}
}	

TP Id	TP_PSAP_SIP_BYE_BV_02
Test Objective	IUT successfully trigger SIP BYE and accepts 200 OK response
Reference	ETSI TS 103 479 [1], clauses 5.4 and 6.1.2.2 Next Generation 112 LTD [i.3], clause 4.7 IETF RFC 3261 [6] IETF RFC 5301 [7]
Configuration	CFG_PSAP_01
PICS Selection	PICS_PSAP_S_SIP_BYE1 and PICS_PSAP_S_SIP_TCP1 and PICS_PSAP_E_SIP_URN1 and PICS_PSAP_B_SDP_ULA1
Initial Conditions	

with {
the IUT inAnActiveIncomingCall
}
Expected Behaviour	

ensure that {
when {
the IUT sends a SIP_BYE
}
then {
the IUT receives a SIP_200OK_BYE
}
}	

7.2.4.4 Message

TP Id	TP_PSAP_SIP_MESSAGE_BV_01
Test Objective	IUT successfully handles an incoming SIP MESSAGE
Reference	ETSI TS 103 479 [1], clauses 5.4 and 6.1.2.2 Next Generation 112 LTD [i.3], clause 4.7 IETF RFC 3261 [6]
Configuration	CFG_PSAP_01
PICS Selection	PICS_PSAP_M_SIP_URN1 and PICS_PSAP_S_SIP_REGISTRATION
Initial Conditions	

with {
the IUT acceptingIM
}
Expected Behaviour	

ensure that {
when {
the IUT receives a SIP_MESSAGE containing
Request_URI indicating value PX_PSAP_SERVICE_URN
 	
}
then {
the IUT sends a SIP_200OK_MESSAGE
}
}	

TP Id	TP_PSAP_SIP_MESSAGE_BV_02
Test Objective	IUT successfully handles an incoming SIP MESSAGE with Call-info and Geolocation header fields
Reference	ETSI TS 103 479 [1], clauses 5.4, 6.1.2.2 and 6.1.2.3 Next Generation 112 LTD [i.3], clause 4.7 IETF RFC 3261 [6]
Configuration	CFG_PSAP_01
PICS Selection	PICS_PSAP_E_SIP_URN2 and PICS_PSAP_S_SIP_REGISTRATION
Initial Conditions	

with {
the IUT acceptingIM
}
Expected Behaviour	

ensure that {
when {
the IUT receives a SIP_MESSAGE containing
Request_URI indicating value PX_PSAP_SERVICE_URN,
Geolocation indicating value PX_GEOLOCATION,
Call_Info indicating value PX_CALL_INFO
}
then {
the IUT sends a SIP_200OK_MESSAGE
}
}	

7.2.4.5 Options

TP Id	TP_PSAP_SIP_OPTIONS_BV_01
Test Objective	IUT successfully handles an incoming SIP OPTIONS
Reference	ETSI TS 103 479 [1], clauses 5.4 and 6.1.2.2 Next Generation 112 LTD [i.3], clause 4.7 IETF RFC 3261 [6]
Configuration	CFG_PSAP_01
PICS Selection	PICS_PSAP_S_SIP_OPT1 and PICS_PSAP_S_SIP_NO_REGISTRATION
Initial Conditions	

with {
the IUT isDisconnected
}
Expected Behaviour	

ensure that {
when {
the IUT receives a SIP_OPTIONS
}
then {
the IUT sends a SIP_200OK_OPTIONS
}
}	

7.2.4.6 Cancel

TP Id	TP_PSAP_SIP_CANCEL_BV_01
Test Objective	Check that IUT successfully handles an incoming SIP CANCEL, release the call with 200 OK CANCEL and 487
Reference	ETSI TS 103 479 [1], clauses 5.4 and 6.1.2.2 Next Generation 112 LTD [i.3], clause 4.7 IETF RFC 3261 [6]
Configuration	CFG_PSAP_01
PICS Selection	PICS_PSAP_S_SIP_UDP1 and PICS_PSAP_A_SIP_BSC1 and PICS_PSAP_B_SDP_ULA1
Initial Conditions	

with {
the IUT acceptingIncomingCalls
}
Expected Behaviour	

ensure that {
when {
the IUT receives a SIP_CANCEL
}
then {
the IUT sends a SIP_200OK_CANCEL and
the IUT sends a SIP_487_INVITE
}
}	

7.2.4.7 Info

TP Id	TP_PSAP_SIP_INFO_BV_01
Test Objective	IUT successfully handles an incoming SIP INFO
Reference	ETSI TS 103 479 [1], clauses 5.4 and 6.1.2.2 Next Generation 112 LTD [i.3], clause 4.7 IETF RFC 3261 [6] IETF RFC 5301 [7]
Configuration	CFG_PSAP_01
PICS Selection	PICS_PSAP_S_SIP_TCP1 and PICS_PSAP_E_SIP_URN1 and PICS_PSAP_B_SDP_ULA1
Initial Conditions	

with {
the IUT inAnActiveIncomingCall
}	

Expected Behaviour	

ensure that {
when {
the IUT receives a SIP_INFO
}
then {
the IUT sends a SIP_200OK_INFO
}
}	

7.2.4.8 Dequeue registration/deregistration

TP Id	TP_PSAP_HTTP_DEQUEUE_REGISTRATION_BV_01
Test Objective	Check that the IUT registers properly to an ESRP Dequeue
Reference	ETSI TS 103 479 [1], clauses 5.2.4 and 6.2.1 ETSI TS 103 698 [11] Next Generation 112 LTD [i.3], clause 4.7 IETF RFC 3261 [6] IETF RFC 5301 [7]
Configuration	CFG_PSAP_03
PICS Selection	PICS_PSAP_S_HTTP_DEQUEUE_REGISTRATION
Initial Conditions	

with {
the IUT isDisconnected
}	

Expected Behaviour	

ensure that {
when {
the IUT isRequestedtoConnect
}
then {
the IUT sends a GET containing
Uri indicating value "/dqregrequest",
Content_type indicating value "application/json",
Body containing
JsonMessage containing
DequeueRegistrationQueueUri indicating value PX_PSAP_DQ_SIP_URL,
DequeueRegistrationDequeuer indicating value PX_PSAP_DQR_SIP_URL,
DequeueRegistrationExpirationTime indicating value PX_PSAP_DQ_EXPIRY_TIME
 	
to a ESRP and
the IUT receives a SIP_SUBSCRIBE containing
To indicating value "{PSAP SIP URI}",
From indicating value "{ESRP SIP URI}",
Body containing
JsonMessage containing
DequeueRegistrationQueueUri indicating value PX_PSAP_DQ_SIP_URL,
DequeueRegistrationDequeuer indicating value PX_PSAP_DQR_SIP_URL,
DequeueRegistrationExpirationTime indicating value PX_PSAP_DQ_EXPIRY_TIME
 	
from a ESRP and
the IUT sends a SIP_NOTIFY containing
To indicating value "{ESRP SIP URI}",	

```
            From indicating value "{PSAP SIP URI}",
            Body containing
                JsonMessage containing
                    DequeueRegistrationQueueUri indicating value PX_PSAP_DQ_SIP_URL,
                    DequeueRegistrationDequeuer indicating value PX_PSAP_DQR_SIP_URL,
                    DequeueRegistrationExpirationTime indicating value PX_PSAP_DQ_EXPIRY_TIME



        to the ESRP entity
    }
}
```

TP Id	TP_PSAP_HTTP_DEQUEUE_REGISTRATION_BV_02
Test Objective	Check that the IUT accepts incoming calls after having registered to an ESRP Dequeue
Reference	ETSI TS 103 479 [1], clauses 5.2.4 and 6.2.1 ETSI TS 103 698 [11] Next Generation 112 LTD [i.3], clause 4.7 IETF RFC 3261 [6] IETF RFC 5301 [7]
Configuration	CFG_PSAP_02
PICS Selection	PICS_PSAP_S_HTTP_DEQUEUE_REGISTRATION
Initial Conditions	

```
with {
the IUT isConnected and
the IUT isRegistered
            }
```

Expected Behaviour

```
ensure that {
    when {
        the IUT receives a SIP_INVITE containing
            Request_URI indicating value PX_PSAP_SERVICE_URN,
            Content_Type indicating value "application/sdp"

    }
    then {
        the IUT establishesIncomingCall
    }
}
```

TP Id	TP_PSAP_HTTP_DEQUEUE_REGISTRATION_BV_03
Test Objective	Check that the IUT deregisters properly from an ESRP Dequeue
Reference	ETSI TS 103 479 [1], clauses 5.2.4 and 6.2.1.2 ETSI TS 103 698 [11] Next Generation 112 LTD [i.3], clause 4.7 IETF RFC 3261 [6] IETF RFC 5301 [7]
Configuration	CFG_PSAP_03
PICS Selection	PICS_PSAP_S_HTTP_DEQUEUE_REGISTRATION
Initial Conditions	

```
with {
the IUT isConnected
            }
```

Expected Behaviour

ensure that {
when {
the IUT isRequestedtoDisconnect
}
then {
the IUT sends a GET containing
Uri indicating value "/dqregrequest",
Content_type indicating value "application/json",
Body containing
JsonMessage containing
DequeueRegistrationQueueUri indicating value PX_PSAP_DQ_SIP_URL,
DequeueRegistrationDequeuer indicating value PX_PSAP_DQR_SIP_URL,
DequeueRegistrationExpirationTime indicating value "0"
 	
to the ESRP entity
}
}

7.2.4.9 HELD

TP Id	TP_PSAP_HELD_BV_01
Test Objective	Check that the IUT sends an HTTP/HELD request when the IUT is requested to get location information
Reference	ETSI TS 103 479 [1], clause 6.5 ETSI TS 103 698 [11] Next Generation 112 LTD [i.3], clause 4.7.3 IETF RFC 3261 [6] IETF RFC 5301 [7]
Configuration	CFG_PSAP_03
PICS Selection	PICS_PSAP_S_HTTP_HELD
Initial Conditions	

with {
the IUT isConnected and
the IUT isRegistered
}	

Expected Behaviour	

ensure that {
when {
the IUT isRequestedToGetLocationInfo
}
then {
the IUT sends an POST containing
Uri indicating value "/location",
Host,
not Accept,
Content_type indicating value "application/held+xmlcharset=utf-8",
Body containing
XmlMessage containing
version indicating value "1.0",
locationRequest containing
namespace indicating value "urn:ietf:params:xml:ns:geopriv:held"
device containing
uri indicating value DEVICE_NUMBER_POINT
 	
to the LIS entity
}
}	

7.2.4.10 LoST

TP Id	TP_PSAP_LOST_BV_01
Test Objective	Check that the IUT sends an HTTP/LOST request when the IUT is requested to get route emergency call
Reference	ETSI TS 103 479 [1], clause 6.4 ETSI TS 103 698 [11] Next Generation 112 LTD [i.3], clause 4.7.2 IETF RFC 3261 [6] IETF RFC 5301 [7]
Configuration	CFG_PSAP_03
PICS Selection	PICS_PSAP_S_HTTP_LOST

Initial Conditions

```
with {
the IUT isConnected and
the IUT isRegistered
            }
```

Expected Behaviour

```
ensure that {
    when {
        the IUT isRequestedToGetRouteEmergencyCalls
    }
    then {
        the IUT sends an POST containing
            Uri indicating value "/service",
            Host,
            not Accept,
            Content_type indicating value "application/lost+xmlcharset=utf-8",
            Body containing
                XmlMessage containing
                    version indicating value "1.0",
                    findService containing
                        namespace indicating value "urn:ietf:params:xml:ns:lost1",
                        location containing
                            profile indicating value "geodetic-2d",
                            id indicating value LOCATION_ID,
                            Point containing
                                namespace indicating value "http://www.opengis.net/gml",
                                srsName indicating value "urn:ogc:def:crs:EPSG::4326",
                                pos indicating value POINT_IN_E_POLICE_SERVICE_BOUNDARY

                        ,
                        service indicating value E_POLICE_SERVICE_URN




        to the ECRF entity
    }
}
```

7.2.4.11 Security

TP Id	TP_PSAP_TLS_BI_01
Test Objective	IUT rejects requests without a certificate
Reference	ETSI TS 103 479 [1], clause 6.1.1 EENA Next Generation 112 LTD [i.3], clause 4.10 IETF RFC 5985 [3]
PICS Selection	PICS_PSAP_TLS_AUTHENTICATION
Initial Conditions	

with {
the IUT isDisconnected
}	

Expected Behaviour	

ensure that {
when {
the IUT receives any SIP_REQUEST containing
not certificate
 	
}
then {
the IUT rejects the Connection
}
}	

7.2.5 BCF

7.2.5.1 Invite

TP Id	TP_BCF_SIP_INVITE_BV_01
Test Objective	IUT forwards well-formated INVITE to PSAP
Reference	ETSI TS 103 479 [1], clause 5.1 ETSI TS 103 698 [11], clause 5.1.1 EENA Next Generation 112 LTD [i.3], clause 4.2
Configuration	CFG_BCF_01
PICS Selection	PICS_BCF_S_SIP_TCP1
Initial Conditions	

with {
the IUT acceptingIncomingCalls
}	

Expected Behaviour	

ensure that {
when {
the IUT receives a TCP SIP_INVITE containing
Request_URI indicating value PX_BCF_SERVICE_URN,
Content_Type indicating value "application/sdp"
 	
}
then {
the IUT sends a TCP SIP INVITE containing
Request_URI indicating value PX_BCF_REQUEST_URI,
Content_Type indicating value "application/sdp"
 	
}
}	

TP Id	TP_BCF_SIP_INVITE_BV_02
Test Objective	IUT adds EES-source to the INVITE Via header
Reference	EENA Next Generation 112 LTD [i.3], clauses 4.2 and 4.2.2
Configuration	CFG_BCF_01
PICS Selection	PICS_BCF_S_SIP_TCP1
Initial Conditions	

with {
the IUT acceptingIncomingCalls
}
Expected Behaviour	

ensure that {
when {
the IUT receives a TCP SIP_INVITE containing
Request_URI indicating value PX_BCF_SERVICE_URN,
Content_Type indicating value "application/sdp"
 	
}
then {
the IUT sends a TCP SIP_INVITE containing
Request_URI indicating value PX_BCF_REQUEST_URI,
Content_Type indicating value "application/sdp",
Via indicating value "PX_IMS_SUT_BCF_IPADDR or PX_IMS_SUT_BCF_HOME_DOMAIN and PX_IMS_SUT_BCF_PORT"
 	
to the PSAP entity
}
}	

TP Id	TP_BCF_SIP_INVITE_BV_03
Test Objective	IUT adds Incident Tracking ID within Call-Info header to the INVITE message.
Reference	ETSI TS 103 479 [1], clauses 5.1.1 and 6.1.2.7 ETSI TS 103 698 [11], clause 5.1.1
Configuration	CFG_BCF_01
PICS Selection	PICS_BCF_S_SIP_TCP1 and PICS_BCF_M_SIP_CALL_INFO
Initial Conditions	

with {
the IUT acceptingIncomingCalls
}
Expected Behaviour	

ensure that {
when {
the IUT receives a TCP SIP_INVITE containing
Request_URI indicating value PX_BCF_SERVICE_URN,
Content_Type indicating value "application/sdp"
 	
}
then {
the IUT sends a TCP SIP_INVITE containing
Request_URI indicating value PX_BCF_REQUEST_URI,
Content_Type indicating value "application/sdp",
Call_Info indicating value PX_CALL_INFO_INCIDENT_TRACKING_ID
 	
to the PSAP entity
}
}	

TP Id	TP_BCF_SIP_INVITE_BV_04
Test Objective	IUT adds Call ID within Call-Info header to the INVITE message.
Reference	ETSI TS 103 479 [1], clauses 5.1.1 and 6.1.2.7 ETSI TS 103 698 [11], clauses 5.1.1 and 6.1.2.7
Configuration	CFG_BCF_01
PICS Selection	PICS_BCF_S_SIP_TCP1 and PICS_BCF_M_SIP_CALL_INFO
Initial Conditions	

with {
the IUT acceptingIncomingCalls
}
Expected Behaviour	

ensure that {
when {
the IUT receives a TCP SIP_INVITE containing
Request_URI indicating value PX_BCF_SERVICE_URN,
Content_Type indicating value "application/sdp"
 	
}
then {
the IUT sends a TCP SIP_INVITE containing
Request_URI indicating value PX_BCF_REQUEST_URI,
Content_Type indicating value "application/sdp",
Call_Info indicating value PX_CALL_INFO_CALL_ID
 	
to the PSAP entity
}
}	

TP Id	TP_BCF_SIP_INVITE_BV_05
Test Objective	IUT adds Source ID within Call-Info header to the INVITE message.
Reference	ETSI TS 103 479 [1], clauses 5.1.1 and 6.1.2.7 ETSI TS 103 698 [11], clause 5.1.1
Configuration	CFG_BCF_01
PICS Selection	PICS_BCF_S_SIP_TCP1 and PICS_BCF_M_SIP_CALL_INFO
Initial Conditions	

with {
the IUT acceptingIncomingCalls
}
Expected Behaviour	

ensure that {
when {
the IUT receives a TCP SIP_INVITE containing
Request_URI indicating value PX_BCF_SERVICE_URN,
Content_Type indicating value "application/sdp"
 	
}
then {
the IUT sends a TCP SIP_INVITE containing
Request_URI indicating value PX_BCF_REQUEST_URI,
Content_Type indicating value "application/sdp",
Call_Info indicating value PX_CALL_INFO_SOURCE_ID
 	
to the PSAP entity
}
}	

7.2.5.2 Message

TP Id	TP_BCF_SIP_MESSAGE_BV_01
Test Objective	IUT forwards well-formated MESSAGE to PSAP
Reference	ETSI TS 103 479 [1], clause 5.1 ETSI TS 103 698 [11], clause 5.1.1 EENA Next Generation 112 LTD [i.3], clause 4.2
Configuration	CFG_BCF_01
PICS Selection	PICS_BCF_S_SIP_TCP1
Initial Conditions	

with {
the IUT acceptingIM
}
Expected Behaviour	

ensure that {
when {
the IUT receives a TCP SIP_MESSAGE containing
Request_URI indicating value PX_BCF_SERVICE_URN,
Content_Type indicating value "text/plain"
 	
}
then {
the IUT sends a TCP SIP MESSAGE containing
Request_URI indicating value PX_BCF_REQUEST_URI,
Content_Type indicating value "text/plain"
 	
}
}	

TP Id	TP_BCF_SIP_MESSAGE_BV_02
Test Objective	IUT adds EES-source to the MESSAGE Via header.
Reference	EENA Next Generation 112 LTD [I,3], clauses 4.2 and 4.2.2
Configuration	CFG_BCF_01
PICS Selection	PICS_BCF_S_SIP_TCP1
Initial Conditions	

with {
the IUT acceptingIM
}
Expected Behaviour	

ensure that {
when {
the IUT receives a TCP SIP_MESSAGE containing
Request_URI indicating value PX_BCF_SERVICE_URN,
Content_Type indicating value "text/plain"
 	
}
then {
the IUT sends a TCP SIP_MESSAGE containing
Request_URI indicating value PX_BCF_REQUEST_URI,
Content_Type indicating value "text/plain",
Via indicating value "PX_IMS_SUT_BCF_IPADDR or PX_IMS_SUT_BCF_HOME_DOMAIN and PX_IMS_SUT_BCF_PORT"
 	
to the PSAP entity
}
}	

TP Id	TP_BCF_SIP_MESSAGE_BV_03
Test Objective	IUT adds Incident Tracking ID within Call-Info header to the MESSAGE message.
Reference	ETSI TS 103 479 [1], clauses 5.1.1 and 6.1.2.7 ETSI TS 103 698 [11], clause 5.1.1
Configuration	CFG_BCF_01
PICS Selection	PICS_BCF_S_SIP_TCP1 and PICS_BCF_M_SIP_CALL_INFO
Initial Conditions	

with {
the IUT acceptingIM
}
Expected Behaviour	

ensure that {
when {
the IUT receives a TCP SIP_MESSAGE containing
Request_URI indicating value PX_BCF_SERVICE_URN,
Content_Type indicating value "text/plain"
 	
}
then {
the IUT sends a TCP SIP_MESSAGE containing
Request_URI indicating value PX_BCF_REQUEST_URI,
Content_Type indicating value "text/plain",
Call_Info indicating value PX_CALL_INFO_INCIDENT_TRACKING_ID
 	
to the PSAP entity
}
}	

TP Id	TP_BCF_SIP_MESSAGE_BV_04
Test Objective	IUT adds Call ID within Call-Info header to the MESSAGE message.
Reference	ETSI TS 103 479 [1], clauses 5.1.1 and 6.1.2.7 ETSI TS 103 698 [11], clauses 5.1.1 and 6.1.2.7
Configuration	CFG_BCF_01
PICS Selection	PICS_BCF_S_SIP_TCP1 and PICS_BCF_M_SIP_CALL_INFO
Initial Conditions	

with {
the IUT acceptingIM
}
Expected Behaviour	

ensure that {
when {
the IUT receives a TCP SIP_MESSAGE containing
Request_URI indicating value PX_BCF_SERVICE_URN,
Content_Type indicating value "text/plain"
 	
}
then {
the IUT sends a TCP SIP_MESSAGE containing
Request_URI indicating value PX_BCF_REQUEST_URI,
Content_Type indicating value "text/plain",
Call_Info indicating value PX_CALL_INFO_CALL_ID
 	
to the PSAP entity
}
}	

TP Id	TP_BCF_SIP_MESSAGE_BV_05
Test Objective	IUT adds Source ID within Call-Info header to the MESSAGE message.
Reference	ETSI TS 103 479 [1], clauses 5.1.1 and 6.1.2.7 ETSI TS 103 698 [11], clause 5.1.1
Configuration	CFG_BCF_01
PICS Selection	PICS_BCF_S_SIP_TCP1 and PICS_BCF_M_SIP_CALL_INFO
Initial Conditions	

with {
the IUT acceptingIM
}
Expected Behaviour	

ensure that {
when {
the IUT receives a TCP SIP_MESSAGE containing
Request_URI indicating value PX_BCF_SERVICE_URN,
Content_Type indicating value "text/plain"
 	
}
then {
the IUT sends a TCP SIP_MESSAGE containing
Request_URI indicating value PX_BCF_REQUEST_URI,
Content_Type indicating value "text/plain",
Call_Info indicating value PX_CALL_INFO_SOURCE_ID
 	
to the PSAP entity
}
}	

7.2.5.3 Security

TP Id	TP_BCF_TLS_BI_01
Test Objective	IUT rejects requests without a certificate
Reference	ETSI TS 103 479 [1], clause 6.1.1 EENA Next Generation 112 LTD [i.3], clause 4.10 IETF RFC 5985 [3]
PICS Selection	PICS_BCF_TLS_AUTHENTICATION
Initial Conditions	

with {
the IUT isDisconnected
}
Expected Behaviour	

ensure that {
when {
the IUT receives any SIP_REQUEST containing
not certificate
 	
}
then {
the IUT rejects the Connection
}
}	

History

Document history		
V1.1.1	January 2020	Publication
V1.2.1	August 2022	Publication